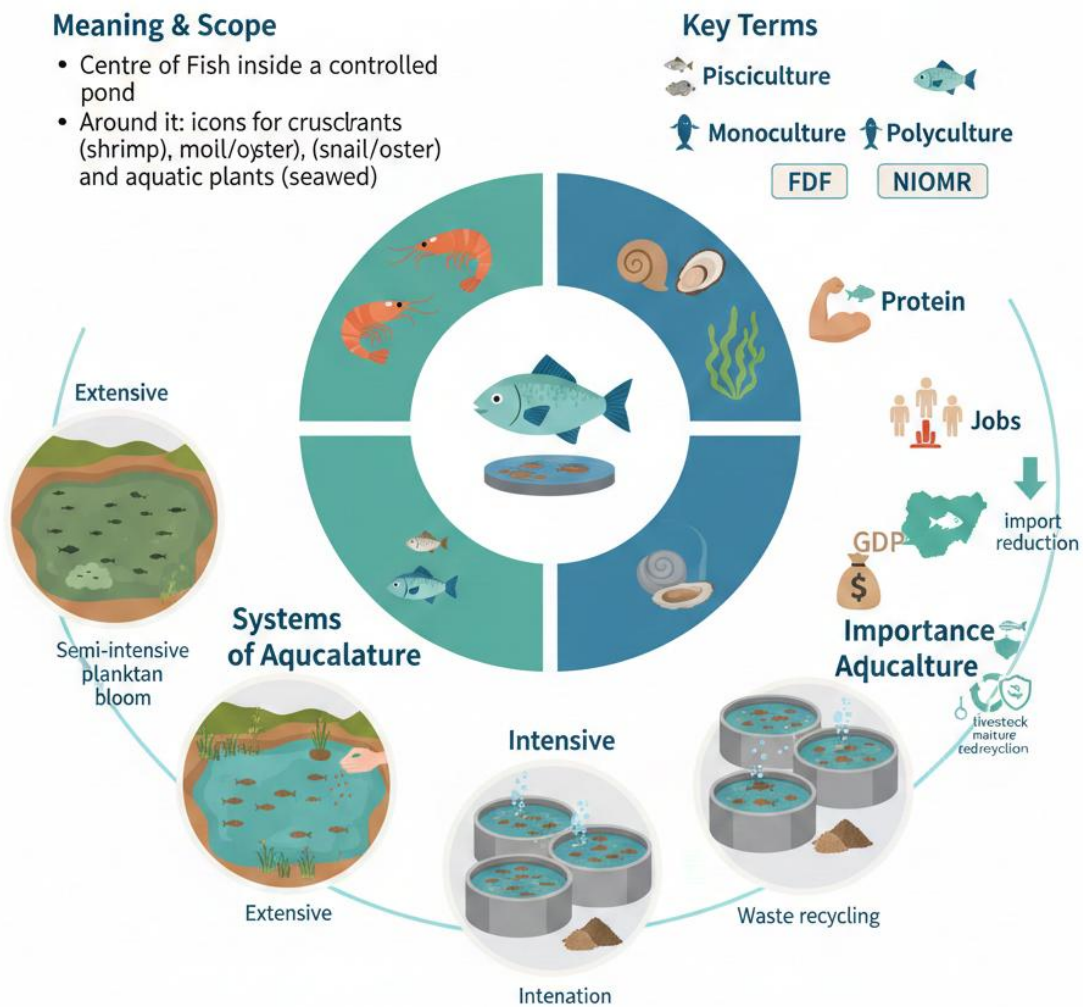


SS2 LIVESTOCK FARMING LESSON NOTES

FIRST TERM

WEEK 1: INTRODUCTION TO AQUACULTURE

INTRODUCTION TO AQUACULTURE



1.1 MEANING AND SCOPE OF AQUACULTURE

Meaning: Aquaculture is the farming of aquatic organisms such as fish, crustaceans (shrimp, crayfish), molluscs (snails, oysters), and aquatic plants (seaweed) in a controlled or semi-controlled environment. It is essentially "farming in water."

It is the direct opposite of **Capture Fishery**, which is the hunting and harvesting of wild fish from natural bodies of water (like oceans, rivers, and lakes). Aquaculture involves human intervention in the breeding, feeding, and rearing processes to enhance production.

Key Terminologies:

- **Pisciculture:** The specific term for fish farming (a branch of aquaculture).
- **Monoculture:** Rearing a single species of fish in a pond (e.g., only Catfish).
- **Polyculture:** Rearing two or more different species (that have different feeding habits) in the same pond (e.g., Tilapia and Catfish).
- **Acronyms:**
 - **FDF:** Federal Department of Fisheries.
 - **NIOMR:** Nigerian Institute for Oceanography and Marine Research.

1.2 IMPORTANCE OF FISH FARMING (AQUACULTURE)

Aquaculture is one of the fastest-growing food production sectors in Nigeria and globally. Its importance cannot be overstated.

1. **Source of High-Quality Protein:** Fish provides excellent animal protein that is rich in essential amino acids, vitamins (D, B12), and minerals (Iron, Iodine). It is often cheaper than beef or chicken.
2. **Employment Generation:** The aquaculture value chain creates millions of jobs. This includes:
 - Direct farm labour (pond construction, feeding).
 - Feed mill operators.
 - Hatchery operators (fingerling production).
 - Fish processors (smokers, freezers).
 - Marketers and transporters.

3. **Economic Growth:** It contributes significantly to the Gross Domestic Product (GDP) of Nigeria. It generates income for farmers and revenue for the government.
4. **Reduction in Fish Imports:** Nigeria has a massive fish demand deficit, spending billions annually to import frozen fish. A strong local aquaculture industry reduces this pressure, saves valuable foreign exchange, and improves national food security.
5. **Conservation of Wild Stocks:** Overfishing (depletion of fish in rivers and oceans) is a global crisis. Aquaculture provides an alternative source of fish, reducing the pressure on wild populations and allowing them to recover.
6. **Utilization of Land and Resources:** Fish farming allows for the productive use of land that may be unsuitable for other forms of agriculture, such as marshy or waterlogged areas.
7. **Waste Recycling:** In integrated farming, waste products from livestock (e.g., poultry droppings, pig manure) can be used to fertilize ponds, promoting plankton growth, which in turn feeds the fish.

1.3 SYSTEMS OF AQUACULTURE

Aquaculture systems are classified based on the level of human intervention, investment, stocking density, and yield.



A. EXTENSIVE SYSTEM

This is the most traditional and low-intensity method.

- **Stocking Density:** Very low (e.g., 1-2 fish per square meter).
- **Pond Size:** Very large, often (e.g., 1-10 hectares).
- **Feeding:** Relies almost 100% on **natural food**. The pond is fertilized (often with manure) to promote the growth of phytoplankton and zooplankton, which the fish eat.
- **Water Management:** Relies on rainfall and natural water bodies.
- **Yield:** Very low (e.g., 300-500 kg/hectare/year).
- **Cost:** Low capital and operating cost.

B. SEMI-INTENSIVE SYSTEM

This is a "middle-ground" approach and is very common in Nigeria.

- **Stocking Density:** Medium (e.g., 5-10 fish per square meter).
- **Pond Size:** Medium (e.g., 0.1 - 1 hectare).
- **Feeding:** Combines **natural food** (from pond fertilization) with **supplementary feed** (e.g., rice bran, groundnut cake, or low-protein farm-made feeds).
- **Water Management:** Some water exchange is required.
- **Yield:** Medium (e.g., 1,500 - 5,000 kg/hectare/year).
- **Cost:** Medium capital and operating cost.

C. INTENSIVE SYSTEM

This is a modern, high-tech, and high-investment method.

- **Stocking Density:** Very high (e.g., 20-100+ fish per square meter).
- **Pond Size:** Small, controlled systems like **concrete tanks, tarpaulin ponds, raceways, or recirculating aquaculture systems (RAS)**.
- **Feeding:** Relies 100% on **complete artificial feed** (high-protein extruded pellets). Fish get no natural food.

- **Water Management:** Requires high-level management, including frequent water exchange, filtration, and **aeration** (pumping air/oxygen into the water) because the high stock density quickly depletes oxygen.
- **Yield:** Very high (e.g., 10,000 - 100,000+ kg/hectare/year).
- **Cost:** Very high capital and operating cost (feed, electricity/fuel for pumping).

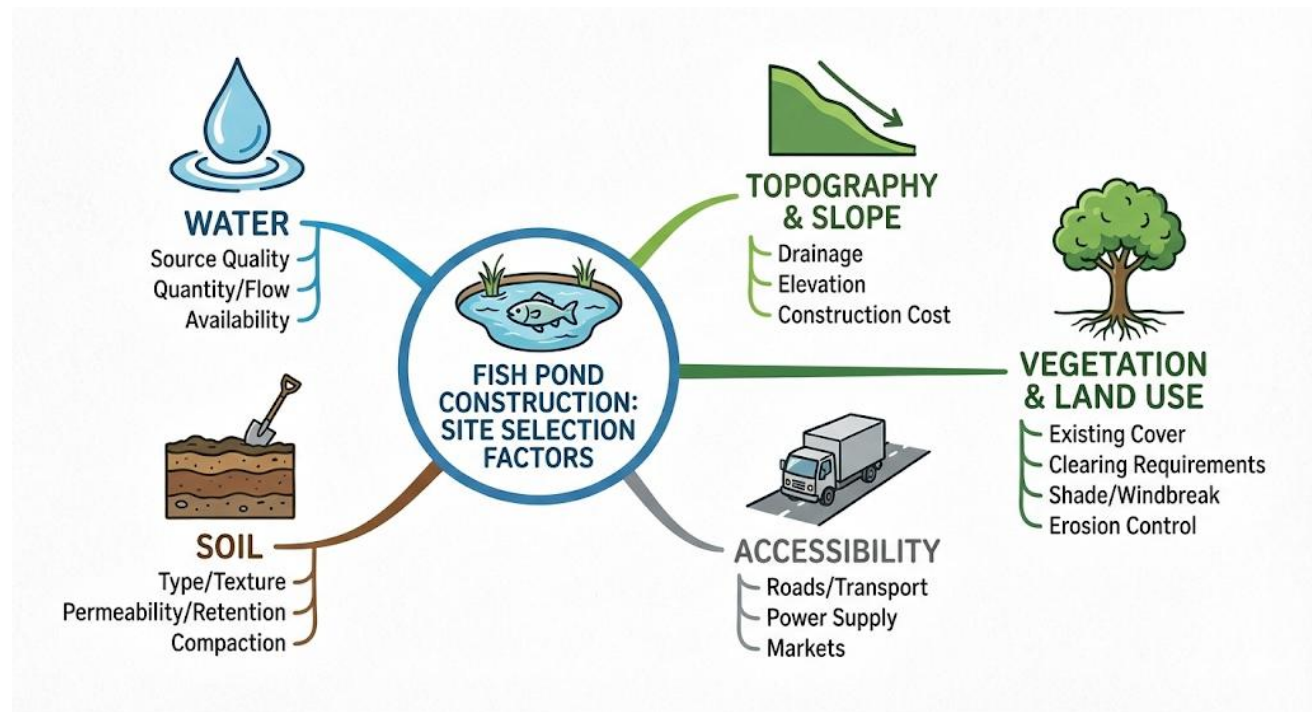
EVALUATION (WEEK 1)

1. Define Aquaculture and differentiate it from Capture Fishery.
2. Explain the difference between Monoculture and Polyculture.
3. List five major importances of fish farming in Nigeria.
4. Compare the Extensive, Semi-Intensive, and Intensive systems using the following criteria:
 - Stocking Density
 - Feeding Method
 - Typical Yield

ASSIGNMENT (WEEK 1)

1. In your own words, write a one-page essay on why the Nigerian government should invest more in aquaculture.
2. Interview a local fish seller (either at the market or at a "point-and-kill" restaurant). Ask them where they get their fish (imported, from the river, or from a farm) and what type of fish is most popular.

WEEK 2: FISH POND CONSTRUCTION



2.1 SITE SELECTION FOR A FISH POND

This is the **most important** decision in fish farming. A bad site will lead to failure, regardless of how well you manage the fish. Key factors to consider:

1. Water Availability:

- **Quantity:** Must be abundant and available all year round. The source (borehole, river, spring, rain) must be reliable.
- **Quality:** The water must be clean and free from pollutants (pesticides, industrial waste, sewage). It should have a neutral pH (6.5-8.5) and good temperature.

2. Soil Type (For Earthen Ponds):

- The soil must be able to **hold water** (low permeability).
- **Ideal Soil:** Clay, clay-loam, or sandy-clay.
- **Bad Soil:** Sandy, gravelly, or rocky soil (will leak water).

- **Field Test:** The "Soil Ball Test": Grab a handful of moist soil, squeeze it into a ball, and toss it slightly. If it holds its shape, it has clay. If it crumbles, it is too sandy.

3. **Topography (Lay of the Land):**

- **Ideal:** A gently sloping area. This is a massive advantage as it allows the pond to be filled and drained entirely by **gravity**, saving energy and money on pumping.
- **Bad:** Very flat land (difficult to drain) or very steep land (expensive to excavate, prone to erosion).

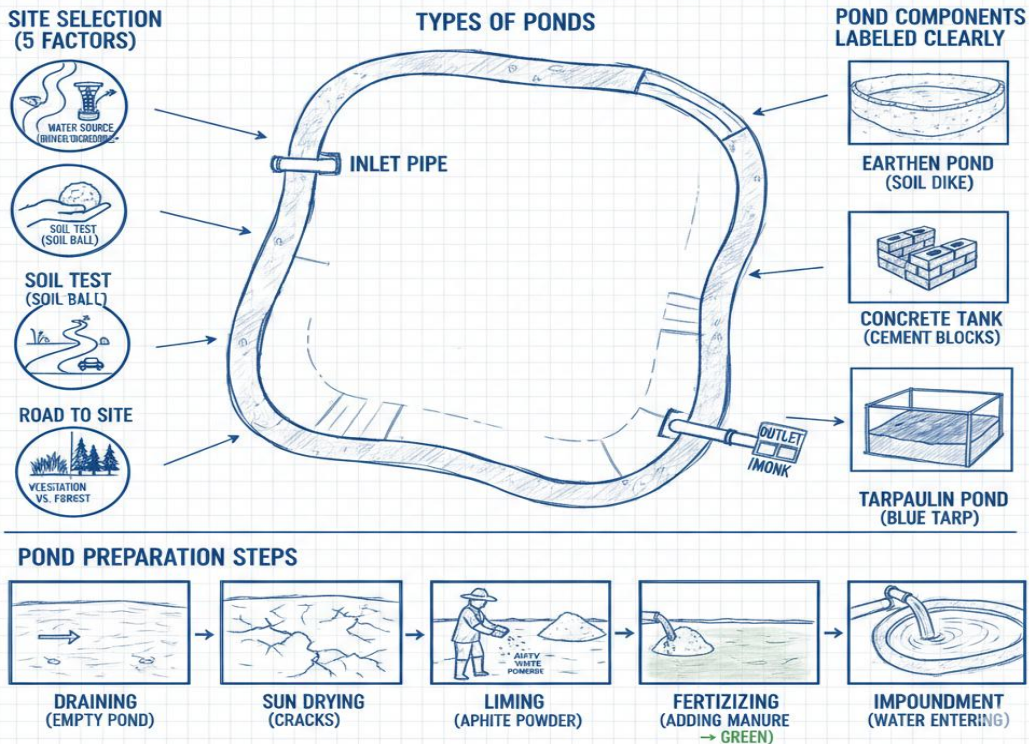
4. **Accessibility:**

- The site must be reachable by road.
- This is crucial for bringing in construction materials, fish feed, and fingerlings, and for transporting harvested fish to the market.

5. **Vegetation and Land Use:**

- A site with minimal vegetation (grass, small shrubs) is cheapest to clear.
- A densely forested area will be very expensive to "stump" (remove tree roots, which can cause leaks).
- Avoid proximity to flood zones or sources of pollution.

THE PERFECT POND BLUEBRUINT



2.2 POND DESIGN AND TYPES

A fish pond is an artificial dam or reservoir built to hold water and fish.

- **Earthen Ponds:** Most common. Formed by excavating soil and using it to build a wall (dike) to impound water.
- **Concrete Ponds/Tanks:** Built with cement blocks and concrete. Common for intensive systems, hatcheries, and in urban areas where land is scarce.
- **Tarpaulin / Plastic Ponds:** A framework (wood or metal) lined with a heavy-duty, waterproof tarpaulin. Popular for beginners and small-scale urban farming.

Components of an Earthen Pond:

- **Dike (Wall):** The compacted wall of earth that holds the water.
- **Pond Bottom:** Must be well-levelled and should slope gently towards the drain.
- **Inlet:** Where water enters the pond.

- **Outlet:** Where water leaves the pond. This is crucial for drainage and harvesting.
- **Spillway:** A channel or pipe that removes excess water (from rain) to prevent the dike from overflowing and breaking.
- **Monk / Sluice Gate:** A common outlet structure that allows the farmer to control the water level and drain the pond from the bottom.

2.3 POND PREPARATION (FOR EARTHEN PONDS)

Before fish are introduced ("stocked"), a new or old pond must be prepared.

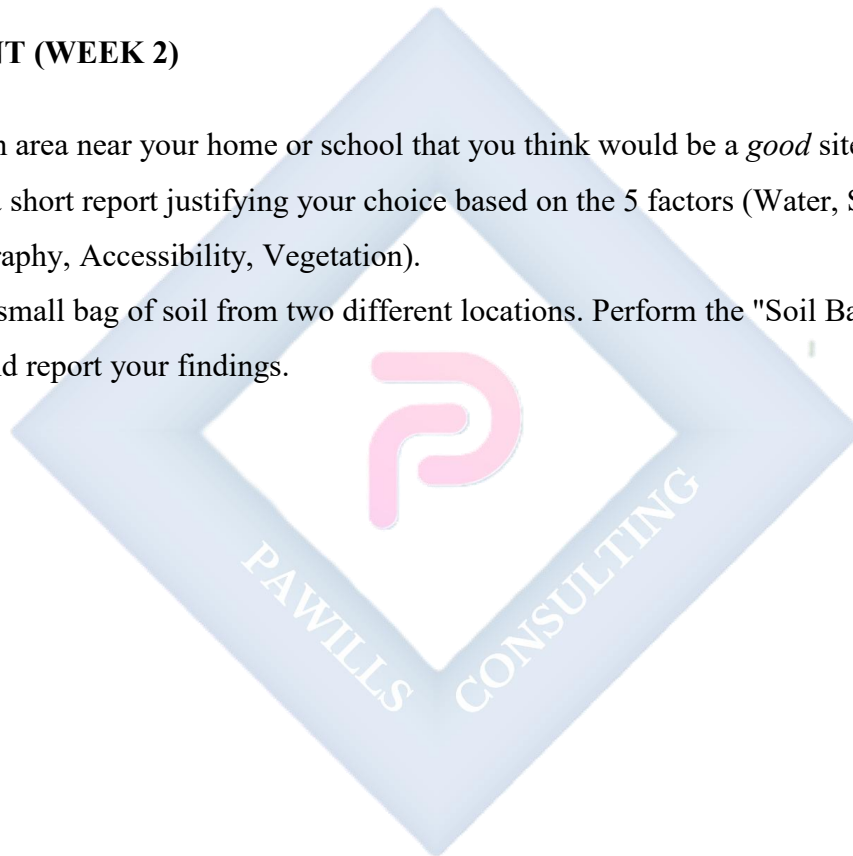
1. **Draining and Drying:** If it's an old pond, drain it completely and allow the bottom to dry in the sun for 1-2 weeks. This kills parasites, predators (like frogs, snakes), and disease organisms.
2. **Liming:**
 - **What:** Applying agricultural lime (Calcium Carbonate, CaCO_3) to the pond bottom.
 - **Why:**
 - To kill any remaining pests and parasites.
 - To neutralize soil acidity and stabilize the water pH.
 - To help clear up muddy water.
3. **Fertilization (The Bloom):**
 - **What:** Adding fertilizers to the water.
 - **Types:**
 - *Organic:* Poultry droppings, pig manure, cow dung.
 - *Inorganic:* Chemical fertilizers (NPK, Urea).
 - **Why:** This is the most critical step for semi-intensive systems. The fertilizer does **not** feed the fish directly. It feeds the **phytoplankton** (microscopic plants), which turn the water green (the "bloom"). The phytoplankton are then eaten by **zooplankton** (microscopic animals). The phytoplankton and zooplankton are the **natural food** for the fish (especially Tilapia).
4. **Impoundment (Filling):** Fill the pond with water. The green, rich "bloom" should appear within 7-10 days. The pond is now fertile and ready for stocking.

EVALUATION (WEEK 2)

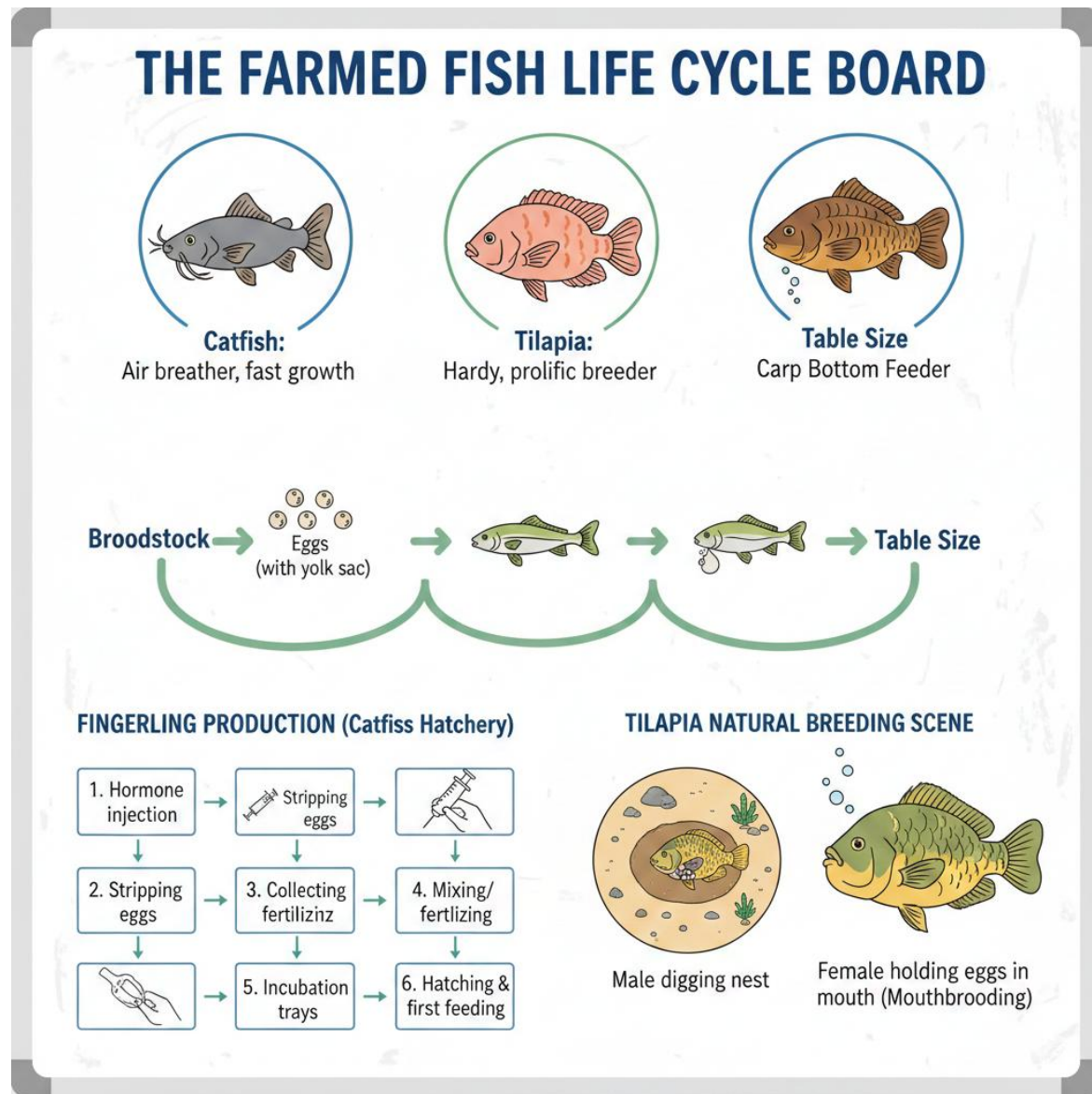
1. List the five most important factors to consider when selecting a site for a fish pond.
2. What is the "Soil Ball Test" used for?
3. Why is a gently sloping topography considered ideal for pond construction?
4. Draw a simple diagram of an earthen pond and label four of its components.
5. Explain the two main reasons for "Liming" a pond.
6. Why do farmers fertilize a pond? What are they trying to grow?

ASSIGNMENT (WEEK 2)

1. Visit an area near your home or school that you think would be a *good* site for a fish pond. Write a short report justifying your choice based on the 5 factors (Water, Soil, Topography, Accessibility, Vegetation).
2. Find a small bag of soil from two different locations. Perform the "Soil Ball Test" on both and report your findings.



WEEK 3: FISH BREEDS AND FINGERLINGS



3.1 COMMON CULTURABLE FISH SPECIES IN NIGERIA

While many fish live in our rivers, only a few are suitable for farming. A "culturable" species must: grow fast, be hardy (tolerate low oxygen), accept artificial feed, have good market demand, and be easy to breed.

1. African Catfish (*Clarias gariepinus* & *Heterobranchus longifilis*)

- This is the **most farmed fish** in Nigeria.

- **Advantages:**
 - **Air Breather:** It has a special "accessory breathing organ" that allows it to breathe atmospheric air. This means it can survive in water with very low oxygen, and thus can be stocked at very high densities.
 - **Hardy:** Very tolerant of poor water quality and stress.
 - **Fast Growth:** Can grow from a fingerling to 1kg (table size) in 4-6 months.
 - **High Market Demand:** The "point-and-kill" and smoked catfish markets are huge.
- **Disadvantage:** They are cannibals (will eat each other if not fed well or sorted by size).

2. Tilapia (*Oreochromis niloticus*)

- This is the second most popular farmed fish.
- **Advantages:**
 - **Hardy:** Tolerant of poor water.
 - **Excellent Natural Feeder:** Feeds low on the food chain (eats plankton and plants), making it cheap to feed in semi-intensive systems.
 - **Good Market Demand:** Has a good, white, flaky flesh.
- **Disadvantage:**
 - **Prolific Breeder:** Tilapia breed *too* easily and *too* often. They can fill a pond with thousands of tiny, unwanted fish. This leads to competition for food and "stunting" (no fish gets big).
 - **Solution:** *Monosex culture* (farming only males, as they grow faster and don't reproduce).

3. Common Carp (*Cyprinus carpio*)

- Less common, but farmed.
- A bottom-feeder, good for polyculture.

3.2 TERMINOLOGY OF FISH STAGES

- **Broodstock:** Mature, healthy, adult male and female fish used for breeding.
- **Egg:** Fertilized fish egg.

- **Fry:** Newly hatched fish. They have a "yolk sac" attached, which provides their food for the first 2-3 days.
- **Fingerling:** The stage after the fry has absorbed its yolk sac and is about the size of a human finger (approx. 5-10 grams). This is the "seed" that farmers buy to stock their ponds.
- **Juvenile:** A young fish, larger than a fingerling but not yet sexually mature.
- **Table Size:** A fish large enough for the market (e.g., 1kg for catfish).

3.3 FINGERLING PRODUCTION (HATCHERY MANAGEMENT)

This is a specialized, technical process of breeding fish to produce fingerlings.

Catfish Breeding (Artificial / Induced Spawning): Catfish will not breed naturally in a small pond. They must be "induced."

1. **Broodstock Selection:** Select a "ripe" female (swollen belly) and a "male" (pointed genital papilla).
2. **Hormone Injection:** The female is injected with a synthetic hormone (e.g., Ovaprim) to trigger **ovulation** (the release of eggs).
3. **Stripping:** After 10-12 hours, the female is ready. The farmer gently presses her belly ("strips" her) to release the eggs into a clean, dry bowl.
4. **Milt Collection:** The male fish is sacrificed (killed), and its testes (milt sac) are removed and cut open.
5. **Fertilization:** The milt (sperm) is squeezed onto the eggs and mixed with a saline solution. The farmer gently stirs for 1-2 minutes.
6. **Incubation:** The fertilized eggs (which are sticky) are spread on spawning nets or in incubation trays with clean, flowing water.
7. **Hatching:** The eggs hatch into fry in 24-36 hours.
8. **First Feeding:** The fry feed on their yolk sacs for 2-3 days, then they are started on a special high-protein feed, such as *Artemia* (brine shrimp).

Tilapia Breeding (Natural Spawning): This is much simpler.

1. Place mature male and female broodstock in a "spawning pond."
2. The male digs a nest, the female lays eggs, and the male fertilizes them.
3. The female (a "mouthbrooder") picks up the eggs in her mouth and holds them until they hatch.
4. The tiny fry are then collected and moved to a "nursery pond."

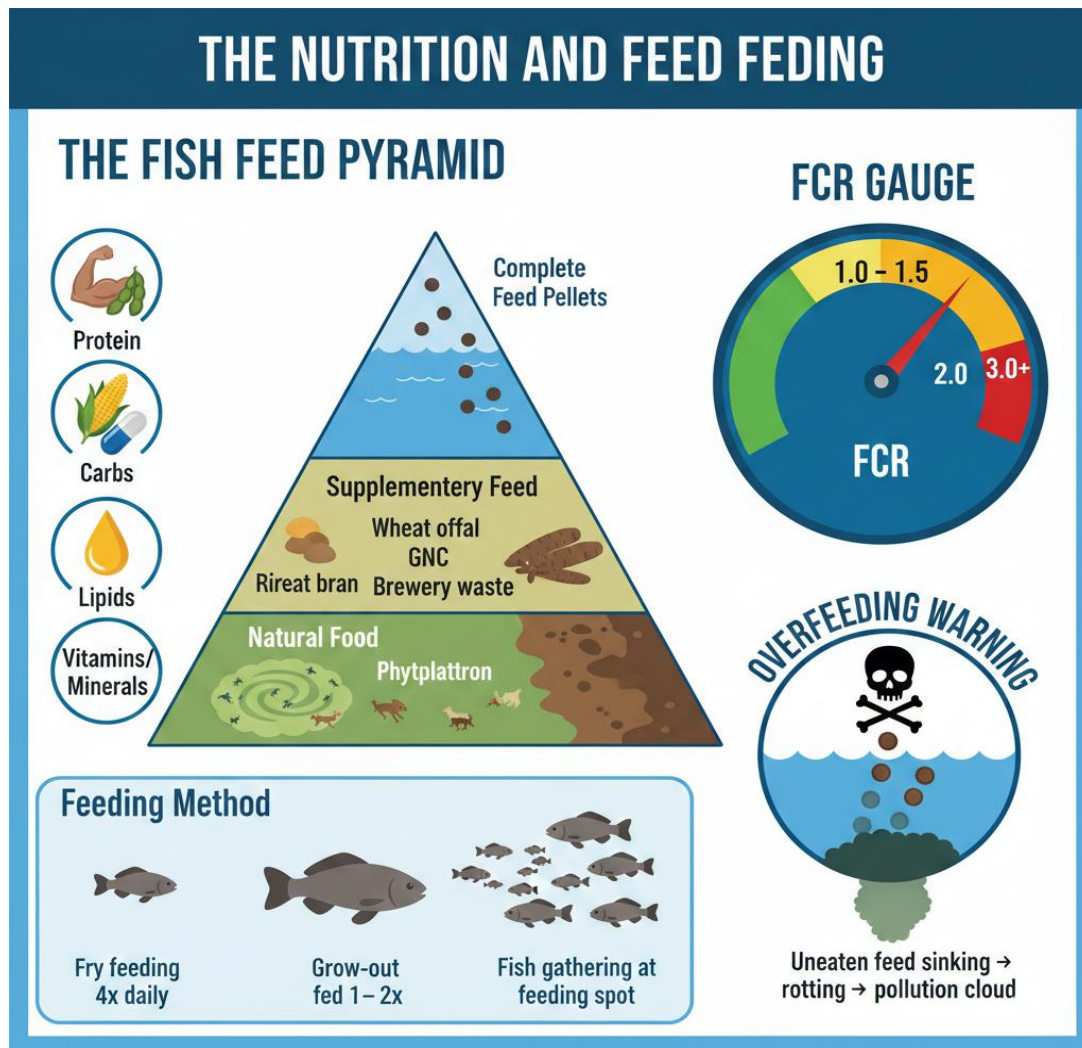
EVALUATION (WEEK 3)

1. State two major advantages and one disadvantage of farming African Catfish.
2. What is the main *disadvantage* of farming Tilapia, and what is the solution?
3. Define the following terms: (a) Broodstock, (b) Fry, (c) Fingerling.
4. Why is a hormone injection *necessary* for breeding catfish in a hatchery?
5. Briefly describe the process of "stripping" and "fertilization" in catfish.

ASSIGNMENT (WEEK 3)

1. Draw a simple flow chart showing the stages of life for a catfish, from Broodstock to Table Size.
2. Research the process of "monosex tilapia" production. Write a short paragraph explaining how farmers create an all-male population.

WEEK 4: FISH NUTRITION AND FEEDING



4.1 IMPORTANCE OF NUTRITION

Fish, like any other animal, require a balanced diet for:

- Growth and weight gain.
- Energy for swimming and metabolic functions.
- A strong immune system to fight disease.
- Reproduction. In intensive systems where no natural food is available, the artificial feed must provide 100% of these needs. **Feed typically accounts for 60-70% of the total operating cost of a fish farm.**

4.2 FISH FEED TYPES

1. Natural Food (in the pond):

- **Phytoplankton:** Microscopic plants (the "green bloom").
- **Zooplankton:** Microscopic animals that eat phytoplankton.
- *Feeds:* Tilapia, Carp, and fry/fingerlings of most species.

2. Supplementary Feed:

- Unprocessed, single-ingredient feeds given to *supplement* natural food in semi-intensive ponds.
- *Examples:* Rice bran, wheat offal, groundnut cake, brewery waste.
- *Problem:* They are not a complete diet and can pollute the water.

3. Complete Artificial Feed (Pellets):

- A carefully formulated and manufactured feed that provides all required nutrients.
- **Pellet Types:**
 - **Floating Pellets:** (Extruded). Most common for catfish. They float on the surface, allowing the farmer to see the fish eat, observe their health, and prevent overfeeding.
 - **Sinking Pellets:** (Compressed). Sink to the bottom. Used for bottom-feeders like shrimp or carp.

4.3 FEED FORMULATION

This is the science of combining different ingredients to achieve a target nutritional profile at the lowest possible cost.

Key Nutrients Required by Fish:

1. Protein (Most Important): The building block for growth.

- Carnivorous fish (like Catfish) need high protein (35-45%).
- Omnivorous fish (like Tilapia) need lower protein (25-35%).
- *Sources:* Fish meal (best, but expensive), Soybean meal, Groundnut cake.

2. Carbohydrates: The primary source of energy.

- *Sources:* Maize (corn), cassava grits, rice bran.

3. **Lipids (Fats/Oils):** A concentrated source of energy.
 - *Sources:* Fish oil, vegetable oil.
4. **Vitamins & Minerals:** Required in tiny amounts for health and metabolic processes.
 - *Source:* "Vitamin Premix" or "Mineral Premix" is added to the feed.

4.4 FEEDING METHODS AND FCR

Feeding Methods:

- **Frequency:**
 - *Fry/Fingerlings (small fish):* Have small stomachs and high metabolism. Must be fed frequently (3-5 times per day).
 - *Grow-out Fish (larger fish):* Fed 1-2 times per day.
- **Timing:** Feed at the same time and same location(s) every day. Fish are creatures of habit.
- **Amount:** The goal is to feed them just enough (to "satiation") without wasting. A common rule is 3-5% of their total body weight per day.

Food Conversion Ratio (FCR):

- **Definition:** FCR is the single most important metric for measuring the efficiency and profitability of a fish farm. It measures how much feed (in kg) is required to produce 1 kg of fish.
- **Formula:**
 - **FCR = Total Weight of Feed Given / Total Weight Gained by Fish**
- **Example:**
 - You give 1,200 kg of feed to your fish over 6 months.
 - You harvest 1,000 kg of fish.
 - $FCR = 1,200 / 1,000 = 1.2$ (This is a good FCR).
- **Interpretation:**
 - A *low* FCR is good (e.g., 1.2:1). It means you are efficient.
 - A *high* FCR is bad (e.g., 3:1). It means you wasted a lot of feed and your profit will be low.
- **Overfeeding is a "double sin":**

1. It wastes money (feed is expensive).
2. The uneaten feed rots at the bottom, polluting the water and causing disease.

EVALUATION (WEEK 4)

1. Why is fish feed so important, and what percentage of operating cost does it represent?
2. What is the main advantage of using "floating pellets" for catfish?
3. List the 4 key nutrients in fish feed and one source for each.
4. Define "Food Conversion Ratio (FCR)".
5. Farmer A uses 1,500 kg of feed to produce 1,000 kg of fish. Farmer B uses 1,000 kg of feed to produce 1,000 kg of fish.
 - (a) What is the FCR for Farmer A?
 - (b) What is the FCR for Farmer B?
 - (c) Which farmer is more efficient?
6. Explain the two major problems caused by overfeeding.

ASSIGNMENT (WEEK 4)

1. Visit a local livestock feed store. Ask for the price of a bag of 6mm Catfish feed. Note the brand and the protein percentage on the bag.
2. Research the "Pearson Square" method. Write a short note on how it is used to formulate a simple fish feed.

WEEK 5: MIDTERM TEST

(This week is for assessment of all topics covered in Weeks 1-4.)

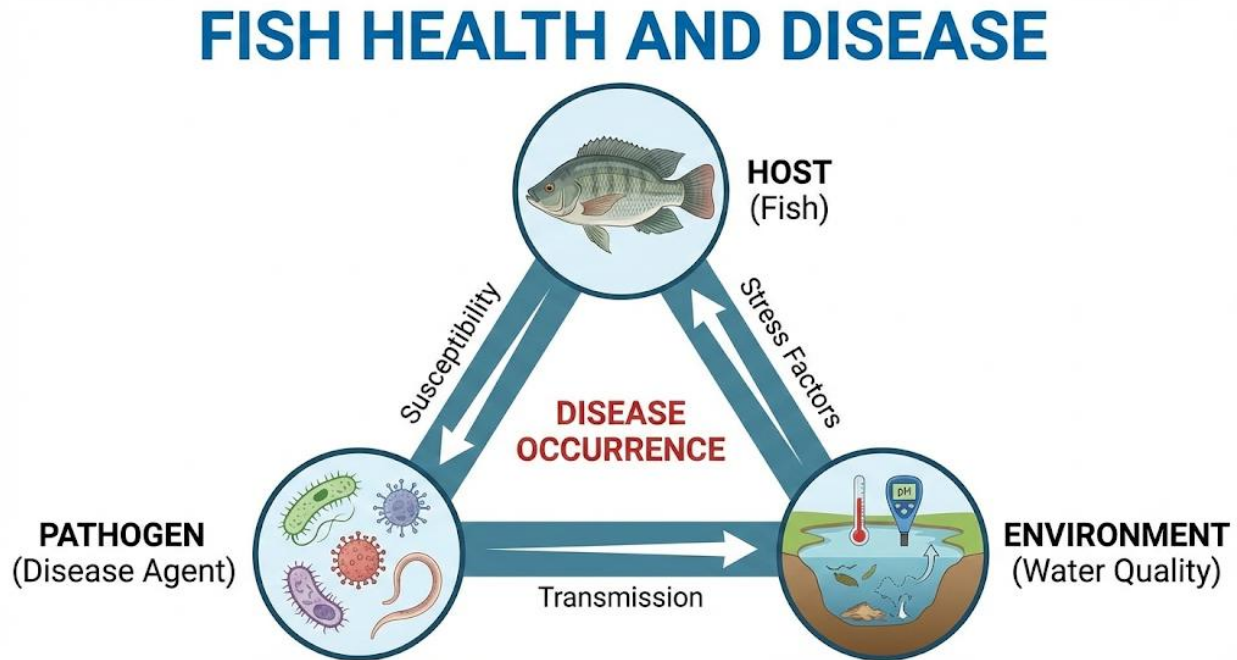


WEEK 6: FISH HEALTH AND DISEASE

6.1 THE DISEASE TRIANGLE

Fish disease is not random. It typically occurs when three factors combine:

1. **A Susceptible Host:** A fish that is *stressed*. Stress is the #1 cause of disease.
 - *Stressors:* Poor water quality (low oxygen), overcrowding, poor nutrition, rough handling.
2. **A Pathogen (Germ):** The bacteria, fungus, or parasite is present in the water.
3. **A Poor Environment:** (See #1 - Poor water quality).



The key to fish health is not killing all pathogens; it's **PREVENTING** stress.

6.2 SIGNS OF A SICK FISH (SYMPTOMS)

A good farmer observes their fish *every day* (especially at feeding time).

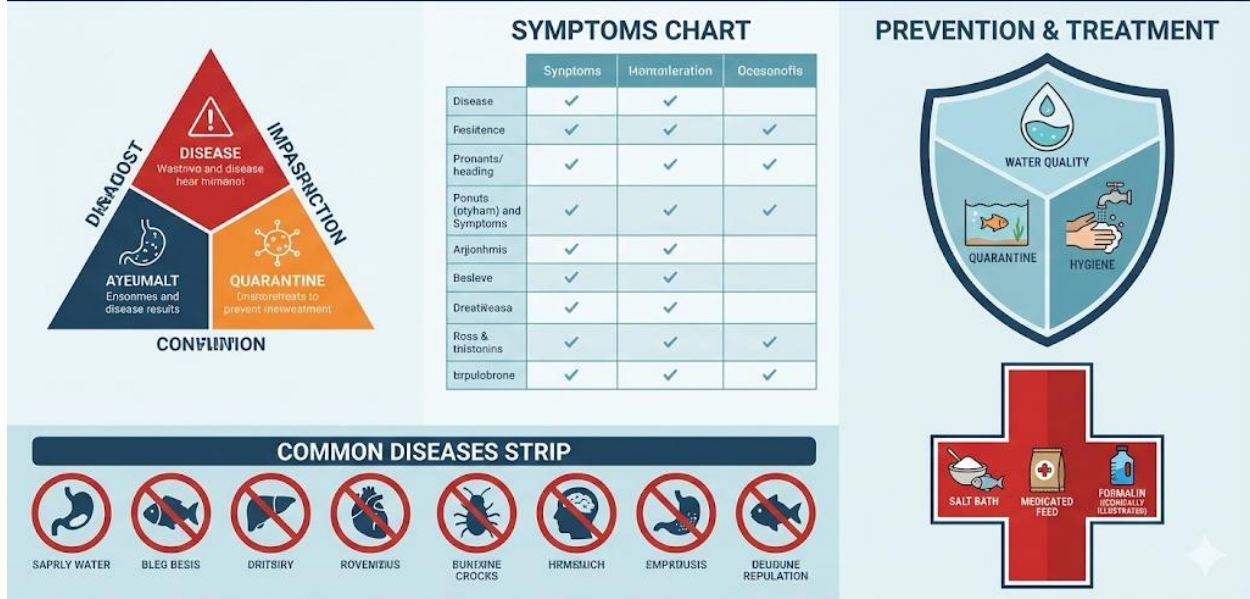
- **Behavioural:**
 - Loss of appetite (not eating).

- Lethargy (weak, floating).
- Gasping for air at the water surface (indicates low oxygen or gill problems).
- "Flashing" or "rubbing" their bodies against the pond walls (indicates skin parasites).
- **Physical:**
 - Frayed or rotting fins.
 - Lesions or red ulcers on the skin.
 - White, cotton-like growths (fungus).
 - Bloated body or "dropsy" (scales stick out like a pine cone).
 - "Pop-eye" (eyes bulging out).
 - High, consistent mortality (e.g., finding 5-10 dead fish every morning).

6.3 COMMON FISH DISEASES

1. **Bacterial Infections:** (e.g., *Aeromonas*, *Columnaris*)
 - *Symptoms:* Fin rot, skin ulcers, red streaks.
 - *Treatment:* Medicated feed (with antibiotics like Oxytetracycline).
2. **Fungal Infections:** (e.g., *Saprolegnia*)
 - *Symptoms:* White, cotton-like growths on skin or eggs.
 - *Note:* Almost always a *secondary* infection that attacks a wound or ulcer.
 - *Treatment:* Salt bath, or Malachite Green.
3. **Parasitic Infections:**
 - *Ich (White Spot Disease):* Caused by *Ichthyophthirius*. Tiny white spots (like salt) on skin and gills.
 - *Costia, Flukes:* Microscopic parasites that cause slimy skin, flashing, and gill damage.
 - *Treatment:* Salt bath, or chemical treatment (e.g., Formalin).

THE BAW DISEASE TRIANGLE



6.4 PREVENTION AND TREATMENT

A. PREVENTION (THE BEST CURE)

- Maintain High Water Quality:** This is the most important rule.
 - Do not overfeed.
 - Ensure proper stocking density.
 - Conduct regular water exchange (flushing).
 - Ensure good aeration (in intensive systems).
- Biosecurity:**
 - Quarantine:** NEVER introduce new fish directly into your main pond. Keep them in a separate "quarantine tank" for 2-3 weeks to watch for signs of disease.
 - Disinfection:** Disinfect all equipment (nets, boots, buckets) before moving between ponds. A simple footbath of salt water or Virkon is effective.
- Good Nutrition:** A well-fed fish has a strong immune system.
- Reduce Handling Stress:** Only handle fish when necessary (e.g., sorting), and do it quickly and gently.

B. TREATMENT

1. **Salt Bath:** The #1 "medicine" in fish farming.
 - **Why:** Salt is cheap, safe, and highly effective. It kills many external parasites and fungi, and it helps the fish by reducing stress on its gills.
 - **How:** Can be a "dip" (high concentration for a short time) or a "bath" (low concentration in the whole pond).
2. **Antibiotics:** Only for bacterial infections. Must be mixed with feed. *Caution: Do not abuse antibiotics.*
3. **Chemicals:** Formalin, Malachite Green. Effective, but dangerous to humans and the fish if overdosed. Use with extreme care.

EVALUATION (WEEK 6)

1. What are the three components of the "Disease Triangle"?
2. What is the single biggest cause of disease in fish, and what are 3 common "stressors"?
3. List 5 symptoms (behavioural or physical) of a sick fish.
4. What is "Biosecurity" in fish farming? Give two examples.
5. What is the most common, cheapest, and safest "medicine" used to treat fish diseases?

ASSIGNMENT (WEEK 6)

1. You observe your catfish "gasping for air at the surface" every morning. What is the *most likely* problem, and what is the immediate solution?
2. Prepare a short report on "Biosecurity." Design a poster (on paper) that a farm could hang up to remind workers of the main rules.

WEEK 7: MIDTERM BREAK

(No new lesson notes for this week.)



WEEK 8: FISH HARVESTING

8.1 WHEN TO HARVEST

Harvesting is done when the fish have reached a desirable "market size" or "table size."

- In Nigeria, this is typically **1kg or more** for catfish ("point-and-kill" size).
- For smoked fish, smaller sizes (300-500g) may be harvested.
- This is determined by the farmer's target market.



8.2 PRE-HARVEST PREPARATION

1. **Stop Feeding:** Stop feeding the fish 1-2 days before the harvest.
 - **Why?** This allows the fish to empty its gut. It reduces stress, prevents the fish from fouling the water with faeces during transport, and improves the taste.
2. **Contact Buyers:** Harvesting is pointless if you have no one to sell to. Confirm your buyers (market women, hotels, processors) and agree on a date and price.
3. **Prepare Equipment:** Get all equipment ready: nets, buckets, basins, weighing scale, ice (if selling fresh), transport tank.

4. **Prepare Labour:** Harvesting is labour-intensive. Organize your workers.

8.3 METHODS OF HARVESTING

A. PARTIAL HARVEST

- **What:** Harvesting only a portion of the fish (usually the largest ones) from the pond.
- **Method:**
 - A **seine net** is used. This is a large net with floats on top and weights on the bottom.
 - The net is dragged from one end of the pond to the other, herding the fish.
 - The "mesh size" of the net is chosen to allow smaller fish to escape, while catching the large ones.
 - The smaller fish are left in the pond to grow bigger for a later harvest.
- **Advantages:** Allows for a continuous source of income.



B. TOTAL HARVEST

- **What:** Harvesting 100% of the fish from the pond. This is common in "all-in, all-out" systems.
- **Method:**

1. The pond is drained slowly through the outlet (Monk).
 2. As the water level drops, the fish concentrate in the deepest part (the "catch basin").
 3. Fish are then easily collected using scoop nets or by hand.
- **Advantages:** Easy to get accurate data on total yield and FCR. Prepares the pond for complete drying, liming, and starting a new cycle.

8.4 PROCESSING OF FISH

Fish is highly perishable (it spoils *very* fast). Processing is done to **preserve** the fish, **add value**, and make it available for longer.

1. Smoking:

- The most popular method of fish preservation in Nigeria.
- The fish is gutted, washed, and often brined (soaked in salt water).
- It is then dried over a hot, smoky fire.
- **Process:** The heat *dries* the fish (removes moisture), and the smoke *coats* it with chemicals that slow bacterial growth and add a desirable flavour.
- *Methods:* Traditional mud kiln, or modern metal smoking ovens.

2. Drying:

- **Sun Drying:** Small fish (like crayfish, Bonga) are spread on mats or racks and dried in the sun.
- **Oven Drying:** Using an oven for a more controlled process.

3. Freezing:

- Requires a "cold chain" - constant electricity for blast freezers and cold storage.
- This is how most imported fish (e.g., "Titus," "Kote") are preserved.

4. Salting:

- The fish is heavily coated in salt, which draws out moisture (osmosis) and kills bacteria.
- Common for stockfish.

EVALUATION (WEEK 8)

1. Why is it important to *stop feeding* fish 1-2 days before harvesting?
2. Differentiate between "Partial Harvest" and "Total Harvest."
3. What is a "seine net," and how is it used?
4. List three common methods of fish processing.
5. Explain the two ways that *smoking* preserves fish.

ASSIGNMENT (WEEK 8)

1. Visit a market and observe the smoked fish section. Ask a seller the price of a medium-sized smoked catfish and a medium-sized frozen mackerel ("Titus"). Compare the prices.
2. Draw a diagram of a seine net being used in a pond.

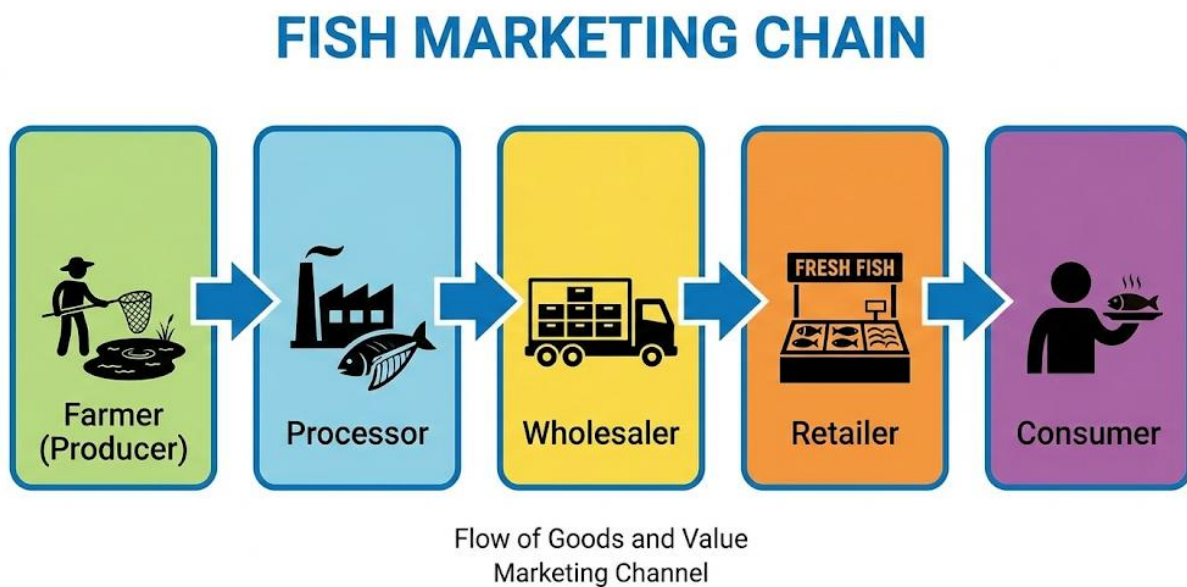


WEEK 9: FISH MARKETING

9.1 THE FISH MARKETING CHAIN

Marketing involves all activities required to move the fish from the farmer to the final consumer.

- **A Simple Chain:**
 - Farmer/Producer → Wholesaler (Market Woman) → Retailer → Consumer
- **A More Complex Chain:**
 - Farmer → Processor (Smoker) → Wholesaler → Retailer → Hotel/Restaurant → Consumer



9.2 STORAGE OF FISH

- **Live Fish:**
 - This is the most profitable way to sell fish ("point-and-kill").
 - Requires holding tanks or drums with clean, aerated water.
- **Fresh (Dead) Fish:**
 - Must be sold *immediately* or preserved in a **cold chain**.

- This means the fish is gutted and placed on **ice** in coolers or ice chests.
- **Processed Fish (Smoked/Dried):**
 - Has a long shelf life (weeks or months).
 - Must be stored in a dry, cool, well-ventilated area.
 - Must be protected from pests (insects, rats) in bags or cartons.

9.3 TRANSPORTATION

This is a critical, high-stress stage.

- **Live Fish Transport:**
 - Fish are loaded into tanks (plastic, fibreglass) or drums on a truck.
 - Water must be clean and cool (transport at night or early morning).
 - **Aeration is essential.** Mechanical aerators (blowers) are used to bubble oxygen into the water.
 - Stocking density must be managed to prevent stress and death.
- **Fresh Fish Transport:**
 - Transported in insulated trucks or coolers filled with ice.
- **Processed Fish Transport:**
 - Can be transported in regular vans or trucks.

9.4 SALES CHANNELS

This is *where* the farmer sells the fish.

1. **Farm Gate Sales:** The buyer comes to the farm to buy the fish. This is the easiest for the farmer as it requires no transport and payment is immediate.
2. **Local Markets:** The farmer transports the fish (live, fresh, or smoked) to the local market to sell.
3. **Caterers (Hotels & Restaurants):** A high-value market. They demand a *consistent* and *reliable* supply of a *specific size* of fish.
4. **Fish Processors:** Companies or individuals who buy fresh fish in bulk for smoking or freezing.

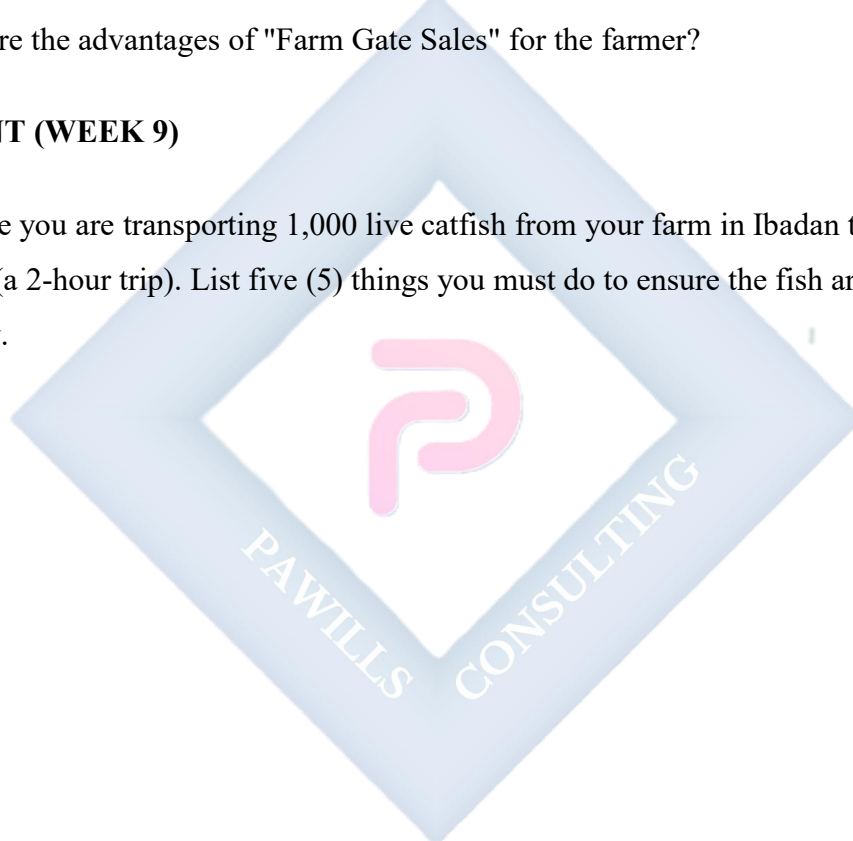
5. **Export:** Selling to other countries. This has very high standards for quality and processing.

EVALUATION (WEEK 9)

1. Draw a simple fish marketing chain with at least 4 links.
2. What is the main requirement for transporting *live* fish?
3. What is a "cold chain" in fish marketing?
4. List four different "sales channels" for a fish farmer.
5. What are the advantages of "Farm Gate Sales" for the farmer?

ASSIGNMENT (WEEK 9)

1. Imagine you are transporting 1,000 live catfish from your farm in Ibadan to a buyer in Lagos (a 2-hour trip). List five (5) things you must do to ensure the fish arrive alive and healthy.

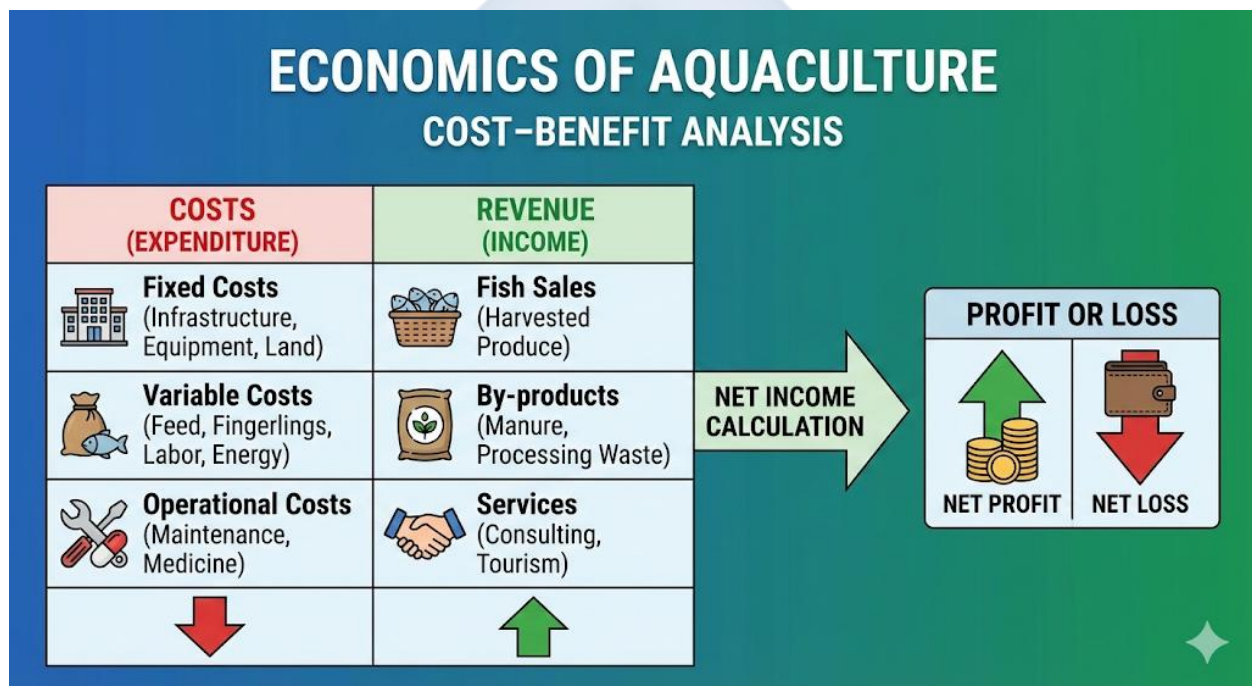


WEEK 10: ECONOMICS OF AQUACULTURE

10.1 FEASIBILITY STUDY

A business plan or "feasibility study" must be done *before* investing any money. It answers the question: "Is this a good idea?" It includes:

- A market survey (Who will buy my fish? At what price?).
- Site analysis (Is my land suitable?).
- Technical analysis (What system will I use? Earthen or concrete?).
- Financial analysis (Costs vs. Benefits).



10.2 COST-BENEFIT ANALYSIS

This is the core of farm economics. **A. COSTS (EXPENDITURE):**

1. **Capital Costs (Fixed Costs):** One-time expenses used to *start* the farm. They do not change.
 - Land (purchase or lease).
 - Pond construction (digging, concrete work).

- Water source (drilling a borehole).
 - Generator, water pump, weighing scale.
 - Vehicle.
2. **Operating Costs (Variable Costs):** The day-to-day expenses of *running* the farm. These change with the number of fish.
- **Fingerlings:** The "seeds."
 - **Feed:** The single **highest** operating cost (60-70%).
 - **Labour:** Salaries for farm workers.
 - **Fuel/Electricity:** For pumping water, aeration, light.
 - **Medication/Salt:**
 - **Transport:**
 - **Repairs:**

B. BENEFIT (REVENUE / INCOME):

- The total amount of money received from selling the fish.
- **Example:** Harvest 950 kg of fish x N1,500/kg = N1,425,000

C. PROFIT:

- **Profit = Total Revenue - Total Operating Costs**
- **Example:** N1,425,000 (Revenue) - N900,000 (Operating Costs) = N525,000 (Profit)

10.3 RECORD KEEPING

This is the most important, but most neglected, part of farm management. If you don't keep records, you are *gambling*, not *farming*. **Why Keep Records?**

- To know if you are making a profit or a loss.
- To calculate your FCR and manage feed costs.
- To spot problems (e.g., "Why is mortality high this month?").
- To make future plans (e.g., "This pond is more productive than that one").
- To get a bank loan (banks demand records).

What to Record:

- **Production Records:** Date stocked, number and weight of fingerlings, daily feed given, daily mortality (deaths), water quality readings, date harvested, total weight harvested.
- **Financial Records:** Every Kobo spent (feed, fuel, labour) and every Kobo earned (sales).

10.4 CHALLENGES OF FISH FARMING IN NIGERIA

1. **High Cost of Feed:** This is the #1 challenge.
2. **Poor Quality Fingerlings:** Many "runts" (stunted fish) are sold to farmers.
3. **Lack of Technical Knowledge:** Many people start without proper training.
4. **Disease Outbreaks:**
5. **Poor Power Supply:** (Affects hatcheries, processing, and cold storage).
6. **Theft / Poaching:** (Human predators).
7. **Access to Credit / Loans:**

EVALUATION (WEEK 10)

1. Differentiate between "Capital Costs" and "Operating Costs." Give two examples of each.
2. What is the single highest operating cost in fish farming?
3. List five reasons why record-keeping is essential for a fish farm.
4. List five major challenges facing fish farmers in Nigeria.
5. A farmer spends N500,000 on feed, N100,000 on fingerlings, and N50,000 on labour. He harvests 1,000 kg of fish and sells them for N800/kg.
 - (a) What is his Total Operating Cost?
 - (b) What is his Total Revenue?
 - (c) What is his Profit or Loss?

ASSIGNMENT (WEEK 10)

1. Design a one-page "Pond Record Sheet" for a farmer to use. It should have columns for: Date, Feed (kg), Mortality (pcs), Water Change (Y/N), and Remarks.

WEEK 11: REVISION

(This week is dedicated to a comprehensive review of all topics from Week 1 to Week 10 in preparation for the examination.)

- **Key Themes for Review:**

- Definitions (Aquaculture, FCR, Monoculture, etc.).
- Systems of Aquaculture (Extensive vs. Intensive).
- Site Selection & Pond Preparation (Why lime? Why fertilize?).
- Fish Species (Catfish vs. Tilapia).
- Fingerling Production (Artificial Spawning).
- Fish Nutrition (Feed formulation, FCR).
- Fish Health (Prevention, Biosecurity, Salt baths).
- Harvesting & Processing (Seine net vs. Draining, Smoking).
- Marketing & Economics (Costs, Records).



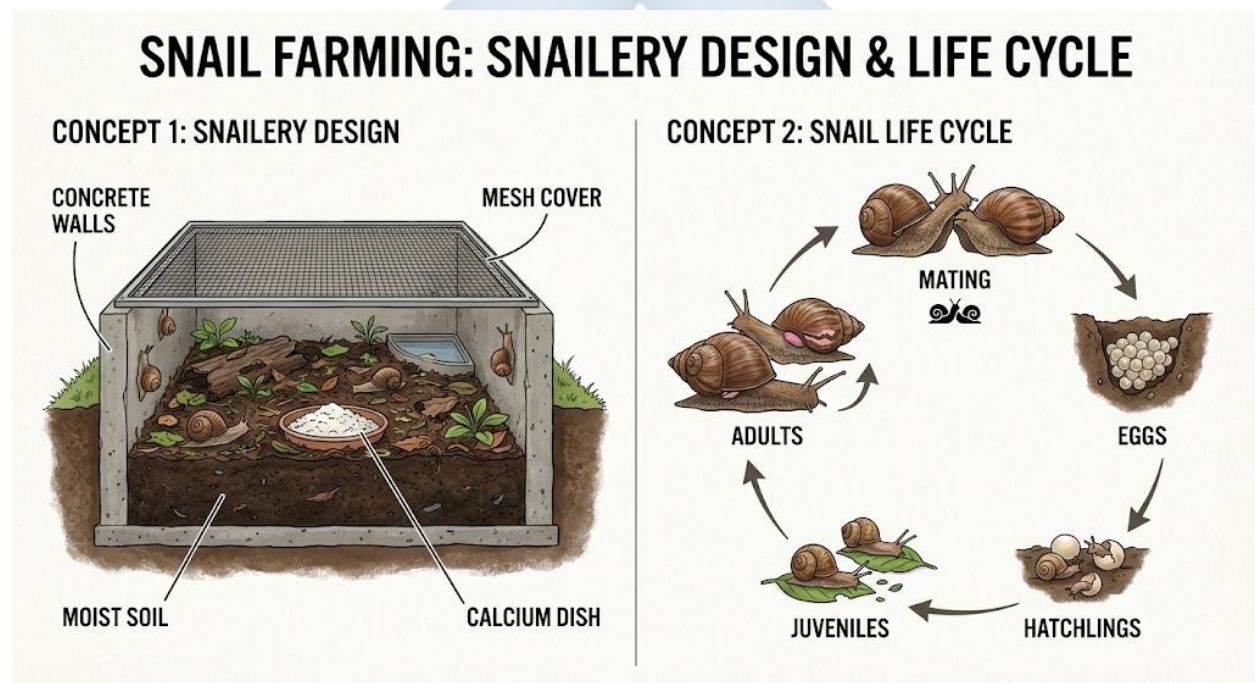
SS2 LIVESTOCK FARMING LESSON NOTES

SECOND TERM

WEEK 1: SNAIL FARMING I (HELICULTURE)

1.1 INTRODUCTION TO SNAIL FARMING (HELICULTURE)

Heliciculture is the specialized branch of agriculture concerned with the farming of land snails for human consumption or other uses. While often overlooked, snail farming is a highly profitable, low-capital, and low-risk venture. The most commonly farmed species in Nigeria is the Giant African Land Snail (specifically *Archachatina marginata* and *Achatina achatina*).



1.2 IMPORTANCE OF SNAIL FARMING

1. **HIGH NUTRITIONAL VALUE:** Snail meat is a high-quality food source.
 - **High Protein:** Rich in protein (approx. 16-18%), which is essential for body-building.

- **Low Fat & Cholesterol:** It is an ideal "white meat" for health-conscious consumers and those with heart conditions.
- **Rich in Minerals:** Contains significant amounts of iron (prevents anaemia), calcium (for bones), and phosphorus.

2. ECONOMIC BENEFITS:

- **High Demand & Price:** Snail meat is a delicacy in Nigeria, commanding a very high price in restaurants (especially "point-and-kill"), hotels, and local markets.
- **Lucrative Export Market:** There is a high demand for large, well-preserved snails in Europe and the Americas.
- **Low Capital Investment:** Snail farming does not require expensive, high-tech equipment or large land areas, making it ideal for beginners.

3. ENVIRONMENTAL BENEFITS:

- **Quiet & Odourless:** Snails are silent and their operations are virtually odourless, meaning farms can be set up in residential areas without disturbance.
- **Waste Converters:** They can be fed on a variety of kitchen waste (fruit and vegetable peels), converting "waste" into valuable protein.

4. OTHER USES:

- **Snail Slime (Mucin):** Highly valued in the global cosmetic industry for skin repair and rejuvenation creams.
- **Snail Shells:** Can be ground into powder and used as a source of calcium (in animal feed) or for crafts and decoration.

1.3 HOUSING AND EQUIPMENT (THE SNAILERY)

The snailery is the house or pen where snails are kept. **Key Principles of Snail Housing:**

- **Escape-Proof:** Snails are excellent climbers and can escape through tiny gaps.
- **Predator-Proof:** Must be secured against pests like rats, snakes, and ants.
- **Humid & Shady:** Must be dark and moist, replicating the snail's natural forest-floor habitat.
- **Good Substrate:** Must have suitable soil for egg-laying and resting.

A. TYPES OF HOUSING SYSTEMS

1. Intensive Systems (Small-scale / Urban):

- **Cages / Hutches:** Raised wooden or metal cages, often stacked, with netting and a soil (substrate) box inside.
- **Plastic Drums / Tanks:** Used plastic drums, cut in half, with drainage holes, soil, and a net cover. Good for beginners.
- **Old Tyres:** Stacking 3-4 old tyres, filling the base with soil, and covering the top with a net.

2. Semi-Intensive Systems (Medium-scale):

- **Snail Pens:** The most common commercial method. A walled "garden" (e.g., 5m x 5m) built with concrete blocks.
- *Key Features:*
 - **Foundation:** A deep concrete foundation or trench (at least 30cm deep) to prevent snails from burrowing out and to stop predators from burrowing in.
 - **Walls:** About 1.5m high, with a smooth plaster finish to make climbing difficult.
 - **Top Cover:** Covered with strong wire mesh or nylon netting to prevent escape and block predators.
 - **Perimeter Lip:** A curved lip or "overhang" on the inside of the wall is often built to further prevent escape.

3. Extensive Systems: A large, fenced-off area of a garden or bush. Not common as it is difficult to manage and find the snails.

B. ESSENTIAL EQUIPMENT

1. Substrate (Soil): This is the "equipment" snails live on. It is critical.

- **Ideal Soil:** Loamy soil, rich in organic matter (humus).
- **Bad Soil:** Clay soil (too hard, poor drainage) or Sandy soil (dries out too fast, poor in nutrients).
- The substrate must be 10-15cm deep to allow snails to burrow and lay eggs.

2. Watering Can: For misting the pen to keep it humid.

3. **Hand Trowel:** For managing the soil.
4. **Feeding Trays:** Shallow wooden or plastic trays. Placing food on trays (not on the soil) prevents contamination.
5. **Hygrometer & Thermometer (Optional but good):** To measure humidity and temperature. (Ideal: 75-90% humidity, 25-30°C).

1.4 FEEDING OF SNAILS

Snails are nocturnal (feed at night) and are herbivores/detritivores.

A. FOOD ITEMS

1. **Leaves:** Cocoyam leaves, pawpaw leaves, amaranth (green), cassava leaves.
2. **Fruits:** Pawpaw (both fruit and leaves are favourites), banana, plantain, mango, watermelon, tomatoes (all are excellent).
3. **Tubers:** Cooked cocoyam, yam, or potato (in small amounts, as they are starchy).
4. **Household Waste:** Rinds from watermelon, plantain peels, pawpaw peels.

B. ESSENTIAL SUPPLEMENT: CALCIUM

- **Why?** Calcium is absolutely **ESSENTIAL** for snails to build and maintain their shells. A lack of calcium leads to thin, brittle shells, poor growth, and poor reproduction.
- **Sources:**
 - Ground eggshells
 - Ground oyster shells
 - Bone meal
 - Limestone powder
 - Cuttlefish bone
- A small dish of one of these calcium sources must **always** be available in the pen.

C. FEEDING MANAGEMENT

- Feed snails in the late afternoon or evening (when they are active).

- Remove all uneaten food the *next morning* to prevent it from rotting and attracting pests like flies, ants, and rats.

EVALUATION (WEEK 1)

1. Define Heliciculture.
2. List three nutritional and three economic importances of snail farming.
3. What is the single most important nutrient supplement for snails, and why is it essential?
4. Describe three key features of a semi-intensive snail pen.
5. Why should a snail farmer feed snails in the evening and remove uneaten food in the morning?

ASSIGNMENT (WEEK 1)

1. Start a small "Snailery Feed" project at home. For one week, collect all suitable fruit and vegetable waste (e.g., pawpaw, plantain, and watermelon peels) from your kitchen.
2. Dry and grind some eggshells. Present this to your teacher as a practical collection of snail feed and calcium supplements.

PAWILLS
CONSULTING

WEEK 2: SNAIL FARMING II

2.1 SNAIL BREEDING

Understanding the snail's life cycle is key to managing a farm.

- **Hermaphrodites:** Snails are hermaphroditic, meaning each snail possesses both male and female reproductive organs.
- **Mating:** Despite this, they are **not** self-fertile. They must mate with another snail to exchange sperm. This cross-fertilization ensures genetic diversity.
- **Egg Laying (Oviposition):** After mating, both snails (now fertile) will burrow into the moist substrate (soil) to lay their eggs. This is why having 10-15cm of good, loose soil is non-negotiable.
- **Eggs:** A snail lays a clutch of 50-200 small, yellowish-white eggs.
- **Incubation:** The eggs incubate in the soil for about 2-4 weeks.
- **Hatchlings:** The tiny, newly hatched snails (with very soft, transparent shells) emerge from the soil. They are very delicate and must be protected. They often eat their own eggshells as their first-ever "calcium meal."

2.2 SNAILERY MANAGEMENT PRACTICES

1. Stocking the Snailery:

- **Source:** Get your foundation stock (breeding snails) from reputable farms, not from the wild (wild snails may be diseased or old).
- **Species:** *Archachatina marginata* (Giant West African Snail) is the most common and robust for farming.
- **Stocking Density:** Do not overcrowd the pen. Overcrowding leads to stress, slow growth, and disease. A good rule is 15-20 mature snails per square meter of pen space.

2. Maintaining Humidity:

- Snails are 80% water and breathe through their skin, which must be moist.
- The pen must be misted/watered *every evening* (or morning in the dry season) to keep humidity at 80-90%. Do not *flood* the pen, just keep it damp.

3. Soil Management:

- The soil (substrate) must be changed every 2-3 months.
- Snails' slime and droppings will build up, making the soil acidic and unsuitable for egg hatching.
- The soil should be turned over (tilled) gently every week to keep it loose (aerated) for egg-laying.

4. Pest and Disease Control:

- **Pests:**
 - **Ants:** The #1 enemy, especially of eggs and hatchlings. *Control:* Use a "gutter" of engine oil or water around the legs of hutch-cages, or a perimeter gutter in pens.
 - **Rats & Snakes:** The most serious predators. *Control:* Ensure pens are secure, with no holes, and have a strong mesh top.
 - **Termites:** Can attack wooden hutches.
- **Diseases:** Snails are very hardy and rarely get sick if hygiene is good. Poor hygiene (rotten food, "dirty" soil) can cause fungal or bacterial infections.

2.3 HARVESTING

- **When to Harvest:** Snails are ready for harvest when they reach sexual maturity, which is visible by a **thickened, hard lip** (the "peristome") on the opening of their shell. This takes about 8-12 months.
- **How to Harvest:**
 - Harvesting is done by hand-picking the mature snails.
 - It's best to harvest at night or in the early morning when the snails are active and visible.
 - Do not harvest all your snails. Always leave the best, largest ones as your future breeding stock.

2.4 MARKETING

- **Forms of Sale:**

- **Live:** Most common. Sold by weight (kg) or by size (e.g., "N10,000 per 20 pieces").
- **Frozen:** Deshelled, cleaned, and frozen. This is common for supermarkets and export.
- **Processed:** Smoked or dried. This adds value and extends shelf life.
- **Marketing Channels:**
 - **Hotels and Restaurants:** The most lucrative market ("point-and-kill"). They need a large, consistent supply.
 - **Local Markets:** Market women buy in bulk from farmers.
 - **Supermarkets:** Buy processed or frozen snails.
 - **Export Market:** A very high-standard market, but extremely profitable.

EVALUATION (WEEK 2)

1. Explain why a snail, which is a hermaphrodite, must still mate.
2. Why is the soil substrate (10-15cm deep) the most critical part of a snail pen?
3. What is the #1 pest of snail hatchlings, and how can it be controlled?
4. How can a farmer identify a snail that is mature and ready for harvest?
5. List three different "forms" in which snails can be marketed.

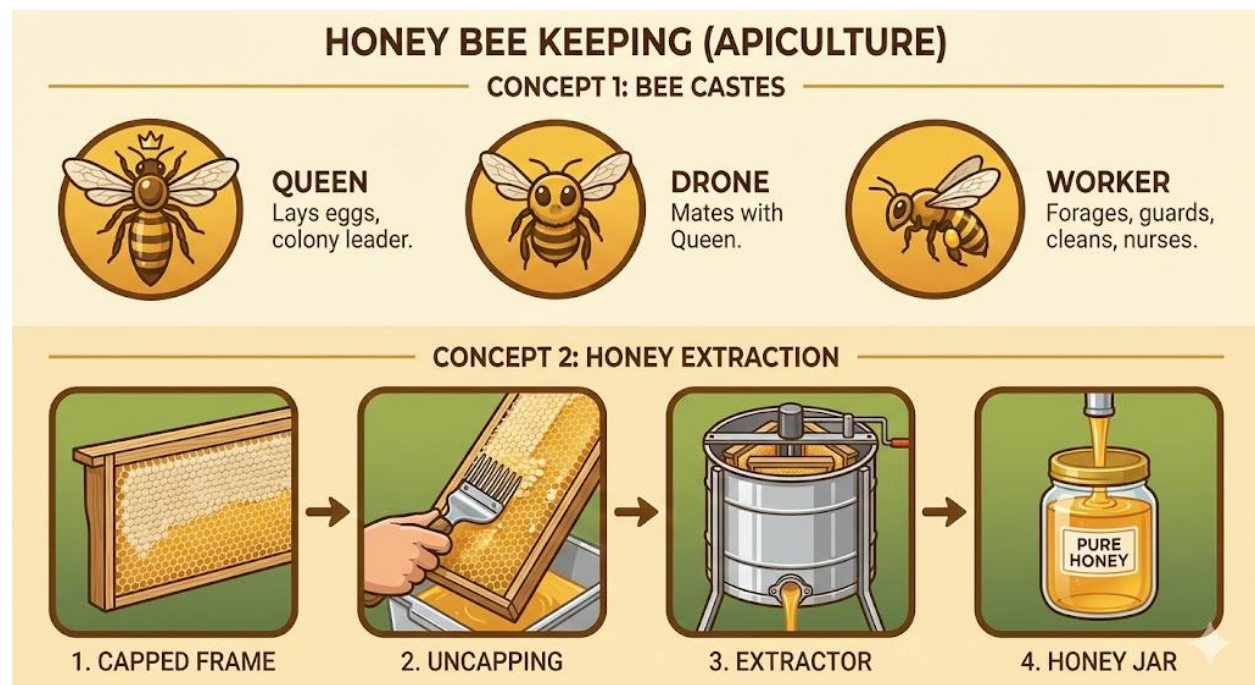
ASSIGNMENT (WEEK 2)

1. Draw a simple diagram of a semi-intensive snail pen. Label five key features, including a pest-control mechanism.
2. Visit a local market or restaurant and ask the price of one mature live snail or a plate of snail meat. Record the price.

WEEK 3: HONEY BEE KEEPING (APICULTURE) I

3.1 INTRODUCTION TO APICULTURE

Apiculture is the science and art of managing honey bees in man-made hives to produce honey and other bee products. It is a vital part of sustainable agriculture.



3.2 IMPORTANCE OF HONEY BEES

Bees are one of the most important animals on the planet.

1. CROP POLLINATION (THE MOST IMPORTANT ROLE):

- Bees are essential for food security. As they forage for nectar, they move pollen from one flower to another, fertilizing the plant.
- This "pollination service" is responsible for increasing the yield and quality of over 70% of our major food crops (e.g., fruits, vegetables, seeds, nuts). A farm with beehives nearby will have a much larger harvest.

2. PRODUCTION OF HONEY:

- Honey is a highly nutritious natural food, rich in sugars (energy) and enzymes.

- It is also a powerful traditional medicine, used for treating wounds, coughs, and infections due to its antibacterial properties.

3. PRODUCTION OF OTHER BEE PRODUCTS (THE HIVE IS A TREASURE CHEST):

- **Beeswax:** Secreted by bees to build the comb. Used in cosmetics (lip balm), candles, and polishes.
- **Propolis ("Bee Glue"):** A resin collected from trees. Bees use it to seal their hive. Humans use it as a powerful antibiotic and antiviral medicine.
- **Royal Jelly:** A special food fed to the queen bee. Used as a high-value nutritional supplement.
- **Pollen:** Trapped from bees' legs. A "superfood" rich in protein.

4. ECONOMIC BENEFITS:

- Provides income for beekeepers (apiarists).
- It is a low-capital venture that can be done on marginal land.

3.3 BEE SPECIES AND CASTES

A. BEE SPECIES

- ***Apis mellifera adansonii*:** The West African Honey Bee. This is our local subspecies. It is well-adapted to our climate and is a good honey producer, but it is known to be very **aggressive** and will "abscond" (abandon the hive) if disturbed.
- ***Apis mellifera*:** The "European Bee." Often imported, tamer, but may be less adapted to our climate.
- ***Meliponini* (Stingless Bees):** Small, non-aggressive bees found in Nigeria. They produce very small amounts of "stingless bee honey," which is highly medicinal and extremely expensive.

B. THE BEE COLONY (SOCIAL CASTES)

A honey bee colony is a "superorganism" with three distinct types (castes) of bees, each with a specific job.

1. THE QUEEN (1 PER HIVE):

- **Function:** She is the *only* reproductive female. Her sole job is to lay eggs (up to 2,000 per day).
- **Characteristics:** Largest bee in the colony, with a long abdomen.

2. THE DRONE (MALE) (Approx. 200-500 per hive):

- **Function:** His only job is to fly out and mate with a new queen from another colony.
- **Characteristics:** Large, fat, with huge eyes. He has no stinger, cannot feed himself, and is expelled from the hive or starves in the dry season.

3. THE WORKER (FEMALE) (Approx. 20,000 - 60,000 per hive):

- **Function:** They are sterile females and do **all** the work in the hive.
- **Characteristics:** Smallest, most numerous.
- **Their jobs change with age:**
 - *Young Bees:* Clean the hive, feed the queen and larvae, build wax combs.
 - *Old Bees:* Guard the hive entrance, fly out to forage (collect nectar, pollen, water, propolis).

3.4 BEEKEEPING EQUIPMENT

1. THE HIVE (THE BEE HOUSE):

- **Traditional Hives:** Clay pots, hollow logs, baskets.
 - *Disadvantage:* Low yield; the colony is often destroyed during harvest (you have to crush the comb).
- **Modern Hives (Movable Frame):**
 - **Langstroth Hive:** The global standard. A system of stacked boxes ("supers") with rectangular, removable frames.
 - **Kenya Top Bar Hive (KTB):** A single-story, long hive with simple "top bars" instead of full frames. It is cheaper and easier to build, making it very popular in African development projects.

2. PROTECTIVE GEAR (PPE):

- **Bee Suit / Overall:** A full-body suit, usually white (bees are less aggressive towards light colours).

- **Veil:** Essential. A mesh screen to protect the face and head.
- **Gloves:** Thick leather gloves to protect the hands.
- **Boots:** Wellingtons, to stop bees from crawling up the legs.

3. HANDLING TOOLS:

- **Smoker:** The most important tool. Puffing *cool, white smoke* into the hive calms the bees (it masks their alarm pheromone and makes them think there is a fire, so they eat honey and become docile).
- **Hive Tool:** A small, steel crowbar used to pry open the hive (which is sealed with propolis) and lift frames.
- **Bee Brush:** A soft brush used to gently move bees off a comb.

EVALUATION (WEEK 3)

1. What is the *most important* role of honey bees in agriculture?
2. List the three "castes" in a bee colony and the main function of each.
3. What is the name of our local, aggressive honey bee subspecies?
4. Why is a "Smoker" the most important handling tool for a beekeeper?
5. What is the main advantage of a modern "Top Bar Hive" over a traditional "Clay Pot Hive"?

ASSIGNMENT (WEEK 3)

1. In your notebook, draw and label the three bee castes (Queen, Drone, Worker), showing their relative sizes.
2. Research "Propolis." Write a short paragraph on what it is and list three medicinal uses.

WEEK 4: HONEY BEE KEEPING II

4.1 APIARY MANAGEMENT

A. THE APIARY

An "Apiary" is the special location or yard where beehives are kept.

B. APIARY SITE SELECTION

This is a critical management decision. A good site leads to a healthy colony; a bad site leads to absconding.

1. **Away from People & Animals:** Hives must be kept far from houses, schools, footpaths, and livestock pens to prevent stinging incidents.
2. **Near Forage (Food):** Must be within 1-3 km of a good source of flowering plants (e.g., citrus orchards, farms, forests).
3. **Near Water:** Bees need water to drink and cool the hive. The site should be near a clean water source (stream, pond, or a provided water tray).
4. **Shade & Windbreak:** Hives should be placed in "dappled" shade (e.g., under a tree) to protect them from the harsh midday sun, which can melt combs. A windbreak (hedge, wall) is also good.
5. **Accessibility:** The site must be accessible for the beekeeper to bring in equipment and carry out heavy honey.
6. **Pest Control:** Hives should be placed on stands (not on the ground) and the legs of the stands placed in tins of oil to prevent ants from climbing up.

C. STOCKING A HIVE (GETTING THE BEES)

1. **Catching a Swarm:** A "swarm" is a natural way bees reproduce. The old queen leaves with half the colony to find a new home. A beekeeper can capture this swarm in a "catch box" and transfer it to their hive.
2. **Baiting a Hive:** An empty hive is set up (baited with beeswax) in a good location during swarming season, and a wild swarm may move in on its own.

D. ROUTINE APIARY INSPECTION

- A beekeeper must inspect the hives every 2-4 weeks.
- **Purpose:**
 - To check if the **Queen is present** and **laying eggs** (seeing eggs is the best sign of a healthy hive).
 - To check for **pests and diseases** (e.g., Wax Moth, Hive Beetle).
 - To check if the bees have enough **space** and **food**.
 - To remove any unwanted "queen cells" to prevent swarming.

4.2 HONEY EXTRACTION AND PROCESSING

A. WHEN TO HARVEST

- Harvesting is done at the end of a major "nectar flow" (the peak of the flowering season).
- You harvest when the bees have filled the combs with honey and "capped" (sealed) at least 80% of the cells with a thin layer of white wax.
- "Capped honey" is "ripe honey." It has the right moisture content (under 18%) and will not ferment.

B. HARVESTING PROCESS

1. **Calm the Bees:** Wear full PPE. Use the smoker at the hive entrance and under the lid to calm the bees.
2. **Remove Combs:** Open the hive and gently brush the bees off the "honeycombs" (the frames full of honey).
3. **Transport:** Place the combs in a secure, bee-proof container to transport them away from the apiary for extraction.
4. **Harvest at Night:** It is often best to harvest at night when all the bees are in the hive and are less aggressive.

C. HONEY EXTRACTION

This is the process of getting the honey *out* of the wax combs.

1. **Crush and Strain Method (For Top Bar Hives / Traditional):**

- The entire comb is cut up and crushed.
- The crushed comb is placed in a clean cloth or sieve.
- The honey drains (strains) through the cloth, leaving the wax behind.
- *Advantage:* Simple, low-cost.
- *Disadvantage:* Destroys the beautiful wax comb, which the bees must work hard to rebuild.

2. **Centrifugal Extraction Method (For Langstroth Hives):**

- **Uncapping:** The wax caps are sliced off the frames using a hot "uncapping knife."
- **Spinning:** The frames are placed in a "Honey Extractor," which is a large drum that spins at high speed.
- Centrifugal force slings the honey out of the cells without destroying the comb.
- **Straining:** The honey is then strained to remove any wax bits.
- *Advantage:* High quality, and the empty combs can be returned to the bees, so they can refill them quickly.

D. PROCESSING AND STORAGE

- **Honey is hygroscopic:** It absorbs moisture from the air, which can cause it to ferment and spoil.
- **Storage:** Honey must be stored in clean, dry, **airtight containers** (e.g., glass jars or food-grade plastic buckets) in a cool, dark place.
- **Crystallization:** Good, pure honey will naturally crystallize (become solid) over time. This is *not* a sign of spoilage. It can be reversed by gently warming the jar in warm water (do not boil).

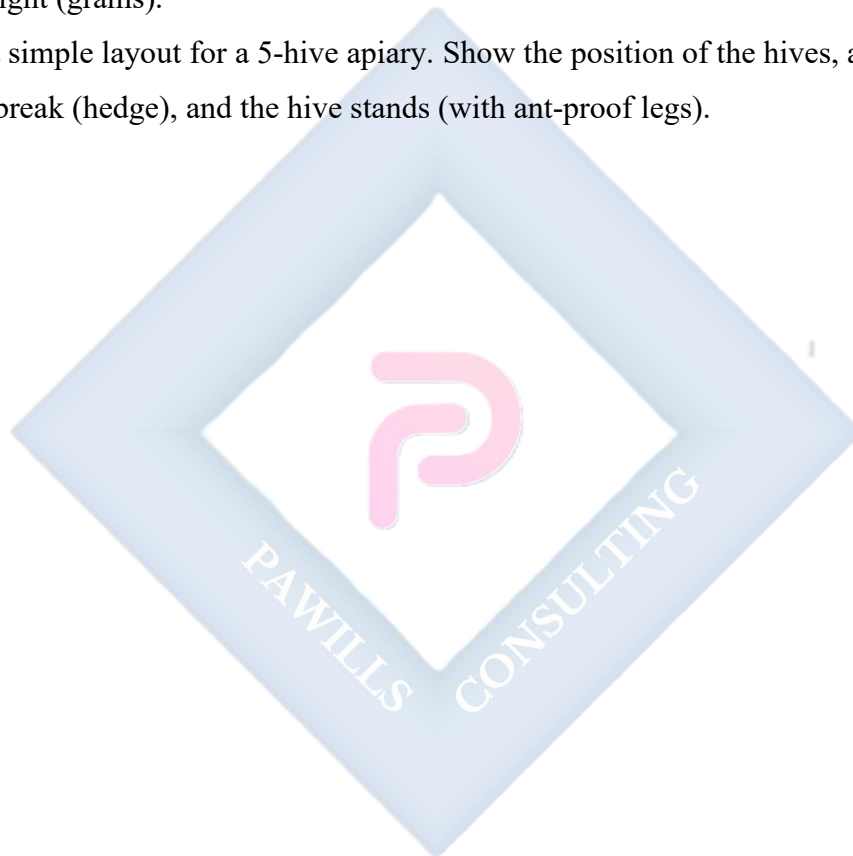
EVALUATION (WEEK 4)

1. What is an "Apiary"?
2. List four key factors to consider when selecting a site for an apiary.

3. What is the difference between the "Crush and Strain" and "Centrifugal Extraction" methods of harvesting honey?
4. What does it mean when honey is "capped," and why is it important for harvesting?
5. Why must honey be stored in an airtight container?

ASSIGNMENT (WEEK 4)

1. Visit a supermarket. Find two different brands of honey. Write down their names, prices, and weight (grams).
2. Draw a simple layout for a 5-hive apiary. Show the position of the hives, a water source, a windbreak (hedge), and the hive stands (with ant-proof legs).



WEEK 5: MIDTERM TEST

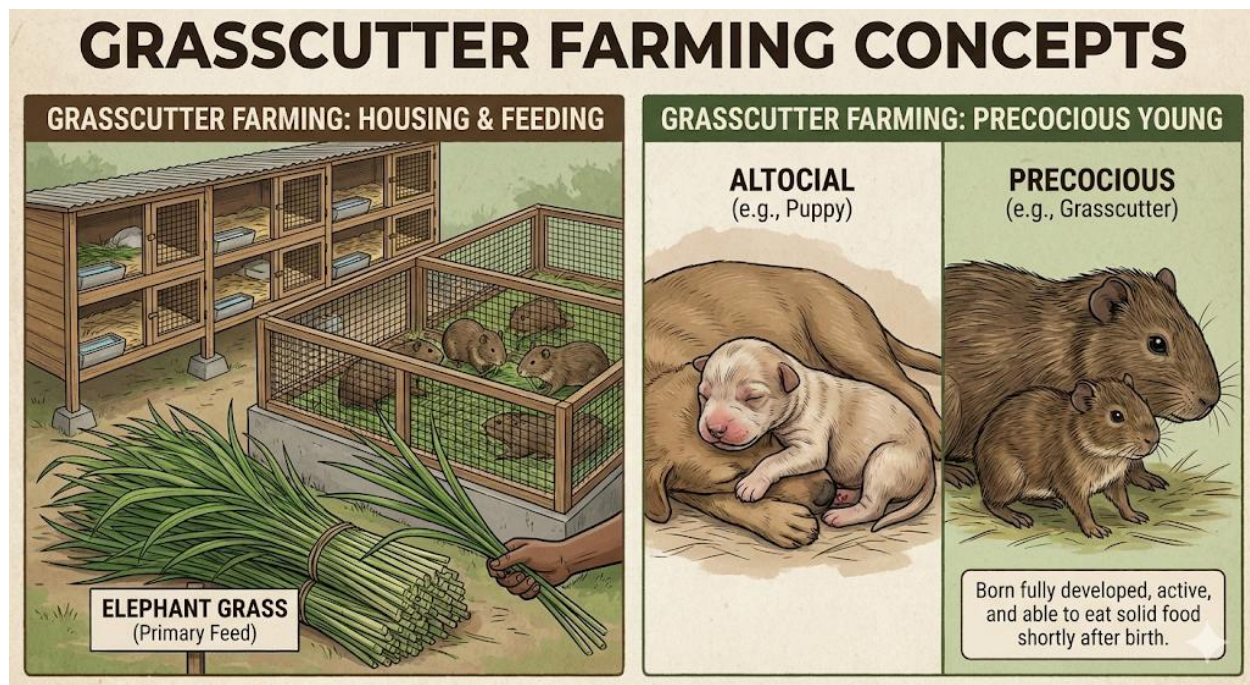
(This week is for assessment of all topics covered in Weeks 1-4: Snail Farming I & II, Honey Bee Keeping I & II.)



WEEK 6: GRASSCUTTER FARMING (THRYONOMY)

6.1 INTRODUCTION

Grasscutter farming (also known as Thryonomy) is the domestication and rearing of the Giant Cane Rat (*Thryonomys swinderianus*). This animal is a large rodent, not related to the common rat, and is a popular "bush meat" in West Africa.



6.2 IMPORTANCE OF GRASSCUTTER FARMING

1. **Economic Value:** Grasscutter meat is a highly sought-after delicacy, commanding a much higher price per kg than chicken, beef, or pork.
2. **Source of Protein:** Provides high-quality animal protein.
3. **Conservation:** Farming grasscutters (domestication) reduces the hunting pressure on wild populations. Wild hunting often involves bush burning, which is environmentally destructive.
4. **Employment:** Creates jobs for farmers, cage-builders, and marketers.
5. **Low Input:** They feed on cheap, readily available grass.

6.3 HOUSING FOR GRASSCUTTERS

- **Key Principle:** The housing must be **gnaw-proof** (they chew everything), secure, cool, and well-ventilated.
- **Hutch System (Like Rabbits):** Raised cages made of metal frames and strong wire mesh.
 - *Advantage:* Easy to clean, saves space.
 - *Disadvantage:* Expensive; wire mesh can cause "Bumble Foot" (sores on the feet).
- **Pen System (Most Common):**
 - A structure built on a concrete floor with low walls (about 1.5m high).
 - The *internal walls must be plastered very smoothly* to prevent the grasscutter from getting a grip to climb or chew.
 - **Compartments:** The pen must be divided into:
 - A "Living Area" (open, for feeding).
 - A "Sleeping Area" (a dark, covered box or "den" inside the pen).
 - They are kept on the floor, which can be covered with wood shavings.

6.4 FEEDING OF GRASSCUTTERS

Grasscutters are strict herbivores.

1. **FORAGE (80% OF DIET):** This is the main food.
 - **Elephant Grass (Napier Grass):** Their absolute favourite.
 - **Guinea Grass:**
 - **Sugarcane:** They love this. It is excellent for energy and for "taming" them.
 - **Feed "Cut-and-Carry":** Grass is cut from the field and carried to the animals.
2. **SUPPLEMENTS (20% OF DIET):**
 - **Tubers:** Cassava, yam, or potato peels (small amounts). *Note:* Cassava must be dried or wilted to reduce cyanide.
 - **Grains/Concentrates:** Maize, sorghum, or rabbit pellets can be given as a supplement, especially to breeding females.
3. **WATER:**

- They get most of their moisture from fresh grass, but clean drinking water must **always** be provided in heavy, "un-tippable" bowls (e.g., clay bowls).

6.5 BREEDING AND MANAGEMENT

- **Mating System (Polygamous):** One male (Buck) is kept with a "colony" of 4-6 females (Does).
- **Maturity:** They are ready to breed at 6-8 months.
- **Gestation Period:** Very long for a rodent (approx. 152 days, or 5 months).
- **Litter Size:** Small (2-4 young).
- **Young (Pups):** Grasscutter pups are **precocious**. This means they are born fully furred, with their eyes open, and they can walk and run within hours of birth (unlike rabbits or rats, which are born blind and naked).
- **Management:**
 - They are shy and easily stressed. The farm must be kept quiet.
 - Handle them gently (grasp by the tail and support the body).
 - Clean pens regularly to prevent disease.

EVALUATION (WEEK 6)

1. What is the scientific name for grasscutter farming?
2. List three major importances of grasscutter farming.
3. What is the *favourite* food of a grasscutter?
4. Describe what "precocious young" means in relation to grasscutters.
5. What is the standard mating ratio for grasscutters?

ASSIGNMENT (WEEK 6)

1. Explain two reasons why farming grasscutters is better for the environment than hunting them.
2. Design a simple floor plan for a 1-Buck, 4-Doe grasscutter colony pen, showing the living area and the sleeping "den."

WEEK 7: MIDTERM BREAK

(No new lesson notes for this week.)



WEEK 8: ANIMAL HEALTH I (COMMON LIVESTOCK PARASITES)

8.1 INTRODUCTION TO PARASITES

A parasite is an organism that lives on (external) or in (internal) another organism (the "host") and causes harm to the host. Parasites are a major cause of economic loss in livestock farming, leading to poor growth, low production (milk, eggs), and death.



8.2 EXTERNAL PARASITES (ECTOPARASITES)

These parasites live *on* the skin, hair, or feathers of the host.

1. TICKS:

- *Hosts:* Cattle, Sheep, Goats, Dogs.
- *Harm:* They are blood-suckers, leading to anaemia, skin damage, and weakness. Most importantly, they are **vectors** that transmit serious diseases (e.g., Redwater fever, Heartwater).

2. MITES:

- *Hosts:* Poultry, Pigs.

- *Harm:* Tiny organisms that burrow into the skin. Cause intense itching, mange, and feather loss. The "Scaly Leg Mite" in chickens is a common example.

3. LICE:

- *Hosts:* Poultry, Cattle.
- *Harm:* They chew on skin and feathers, causing irritation, restlessness, and reduced feeding.

4. FLEAS:

- *Hosts:* Dogs, Goats, Pigs.
- *Harm:* Blood-suckers that cause itching and anaemia.

Symptoms of Ectoparasites:

- Visible scratching, rubbing against walls.
- Restlessness and irritation.
- Visible pests on the skin.
- Feather loss, hair loss (alopecia), skin sores.

Prevention and Control:

- **Dipping:** Walking animals (especially cattle) through a bath of chemical (Acaricide).
- **Spraying:** Using backpack sprayers to apply insecticide/acaricide to animals and their housing.
- **Dusting:** Applying insecticide powder to poultry.
- **Hygiene:** Keeping pens clean.

8.3 INTERNAL PARASITES (ENDOPARASITES)

These parasites live *inside* the host's body (e.g., intestines, lungs, liver).

1. ROUNDWORMS (NEMATODES):

- *Hosts:* Poultry, Pigs, Cattle.
- *Harm:* Live in the intestines, compete for the host's food, causing malnutrition, "pot belly," and gut blockage.

2. TAPEWORMS (CESTODES):

- *Hosts:* Cattle, Dogs, Sheep.
- *Harm:* Long, segmented worms that live in the intestine and absorb digested food.

3. LIVER FLUKES:

- *Hosts:* Cattle, Sheep, Goats.
- *Harm:* Live in the liver and bile ducts, causing severe organ damage (liver rot).
Note: They have a complex life cycle that requires a **freshwater snail** as an intermediate host.

4. COCCIDIOSIS (A PROTOZOA):

- *Hosts:* Poultry (chickens).
- *Harm:* A microscopic parasite that destroys the gut lining, causing bloody diarrhoea, high mortality in chicks.

Symptoms of Endoparasites:

- Diarrhoea (scours), sometimes with blood.
- Weight loss, poor growth.
- Rough hair coat or ruffled feathers.
- Anaemia (pale gums and eyelids, e.g., "bottle jaw").
- Visible worms in feces.

Prevention and Control:

- **Deworming:** Regular, strategic use of anthelmintic drugs (dewormers) to kill the worms.
- **Rotational Grazing:** Moving animals to a new, "clean" pasture regularly. This breaks the parasite life cycle, as the eggs on the old pasture die off.
- **Sanitation:** Cleaning pens, providing clean food and water (not contaminated by feces).
- **Snail Control:** Fencing off wet areas to control snails (for Liver Fluke).

EVALUATION (WEEK 8)

1. What is the difference between an ectoparasite and an endoparasite?
2. Give two examples of each type.

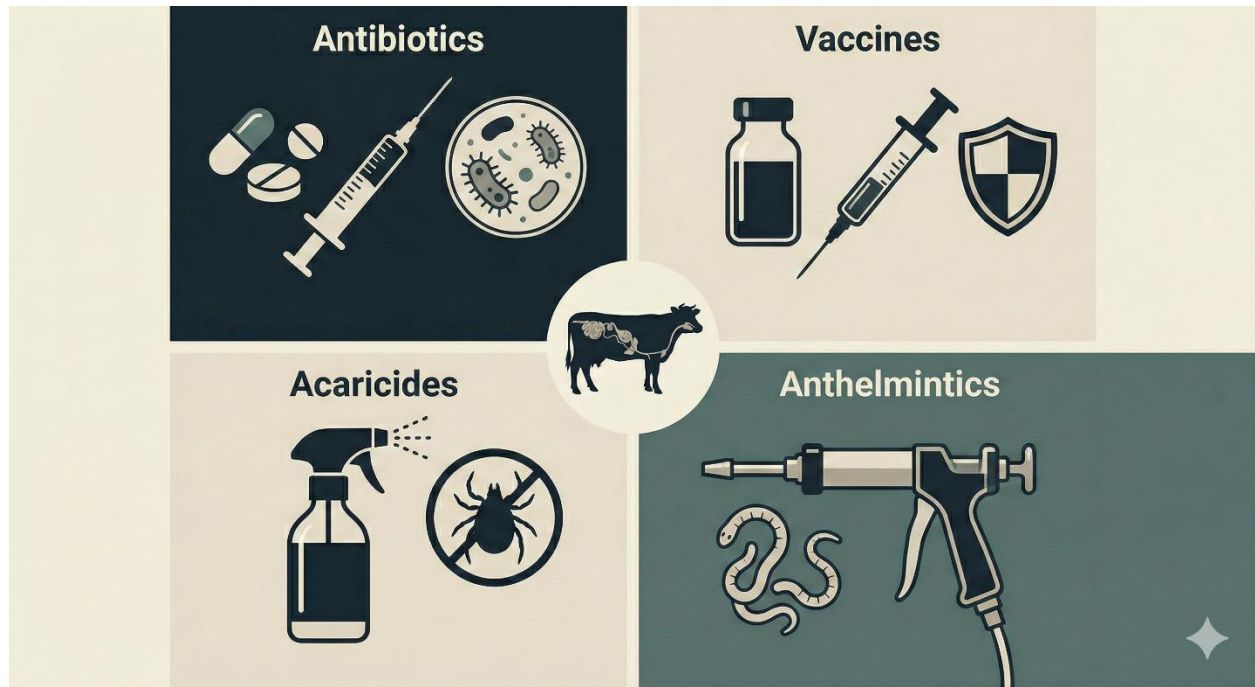
3. What is the main danger of ticks, besides sucking blood?
4. What is "Rotational Grazing," and how does it help control internal parasites?
5. What is the main symptom of Coccidiosis in chicks?

ASSIGNMENT (WEEK 8)

1. Draw the life cycle of the Liver Fluke, making sure to include the intermediate host (the snail).
2. Ask a local poultry or goat farmer what "dewormer" (anthelmintic) they use. Write down the name of the drug.



WEEK 9: ANIMAL HEALTH II (DRUGS, VACCINATION, BIOSECURITY)



9.1 VETERINARY DRUGS AND THEIR USES

1. ANTHELMINTICS (DEWORMERS):

- *Use:* To treat and control **internal parasites** (worms) like roundworms and flukes.
- *Examples:* Albendazole, Ivermectin, Levamisole.

2. ACARICIDES / INSECTICIDES:

- *Use:* To treat and control **external parasites** (ticks, mites, lice).
- *Examples:* Cypermethrin, Amitraz (used in dips/sprays).

3. ANTIBIOTICS:

- *Use:* To treat diseases caused by **BACTERIA** (e.g., pneumonia, scours, wound infections).
- *Examples:* Oxytetracycline (Terramycin), Penicillin.
- *Critical Note:* Antibiotics do **NOT** treat diseases caused by viruses.

4. ANTIPROTOZOALS:

- *Use:* To treat diseases caused by protozoa.

- *Example:* Coccidiostats (e.g., Amprolium) are used to treat or prevent Coccidiosis in poultry.

5. VITAMINS AND SUPPLEMENTS:

- *Use:* To boost the animal's immune system and recovery, especially during periods of stress (e.g., after transport, vaccination, or antibiotic treatment).

9.2 VACCINATION PROGRAMMES

- **What is a Vaccine?** A vaccine is a biological substance (containing a dead or weakened version of a pathogen) that is given to a healthy animal to stimulate its immune system to produce antibodies.
- **Purpose:** It is **PREVENTIVE**, not a cure. It "trains" the body to fight off a *specific* disease if it encounters it in the future.
- **Importance:** This is the cheapest and most effective way to control highly contagious **VIRAL** diseases (like Newcastle Disease), which have no cure.

Example Vaccination Programme for Poultry (Chickens):

- **Day 1:** Marek's Disease (Injection at hatchery).
- **Day 7-10:** Gumboro (IBD) - 1st Dose (Usually in drinking water).
- **Day 14:** Newcastle Disease (ND) - 1st Dose (Lasota strain, in water or eye drop).
- **Day 21:** Gumboro (IBD) - 2nd Dose.
- **Day 28:** Newcastle Disease (ND) - 2nd Dose.
- **Week 6:** Fowl Pox (Wing-web stab). (*This schedule varies, but it shows the principle of regular, timed vaccinations.*)

9.3 QUARANTINE AND BIOSECURITY

This is the "border control" for your farm. Biosecurity refers to all measures taken to stop diseases from *entering* or *spreading* on a farm.

- **QUARANTINE:**

- **Definition:** The isolation of new animals (bought from the market or another farm) *before* they are allowed to join the main flock or herd.
- **Process:** The new animals are kept in a separate pen, far from the main animals, for 2-4 weeks.
- **Purpose:** This "waiting period" allows the farmer to observe the new animals for any signs of disease that they might be incubating. It is the #1 way to prevent



introducing new diseases to your farm.

OTHER BIOSECURITY PRACTICES:

1. **Footbaths:** A shallow pan containing disinfectant placed at the entrance of the farm or each pen. All visitors must step in it to kill germs on their boots.
2. **Limit Visitors:** Do not allow unnecessary people (especially other farmers or marketers) to enter your pens.

3. **All-in, All-out:** A system (common in poultry) where all birds are brought in at once, and all are sold at once. The pen is then completely cleaned and disinfected before the next batch arrives. This breaks the disease cycle.
4. **Cleanliness:** Regular cleaning of feeders, drinkers, and pens.
5. **Pest Control:** Keep rats, wild birds, and flies away, as they can all transmit diseases.

EVALUATION (WEEK 9)

1. What is the difference between an Antibiotic and an Anthelmintic?
2. What is a Vaccine? Is it a preventive measure or a cure?
3. Why is vaccination so important for controlling viral diseases like Newcastle?
4. Define "Quarantine" in livestock farming and explain its purpose.
5. List three other biosecurity measures (besides quarantine) a farmer should use.

ASSIGNMENT (WEEK 9)

1. Design a "Farm Biosecurity" poster. It should have at least 4 rules for visitors (e.g., "STOP: Do Not Enter," "Use Footbath").
2. Research "Newcastle Disease" (ND) in poultry. Write a short paragraph on its symptoms.

WEEK 10: PASTURE AND FORAGE CROPS

10.1 INTRODUCTION

- **Forage:** Any plant material (leaves, stems) grown to be fed to livestock.
- **Pasture:** The land on which forage crops are grown for animals to graze.
- **Ruminants:** These crops are the primary food source for Ruminants (Cattle, Sheep, Goats) and other herbivores (like Grasscutters).
- A good pasture is the cheapest and most effective way to feed ruminant animals.



10.2 TYPES OF PASTURE (FORAGE CROPS)

A balanced diet for a ruminant requires both energy and protein.

- Grasses provide **Energy (Carbohydrates)**.
- Legumes provide **Protein**. An ideal pasture is a *mixture* of grasses and legumes.

A. FORAGE GRASSES (FOR ENERGY)

1. **Elephant Grass (Napier Grass):** (*Pennisetum purpureum*)

- High yield, fast-growing. Propagated by stems (cuttings).
 - Used in a "Cut-and-Carry" system (cut and brought to the animal).
 - Main food for grasscutters.
2. **Guinea Grass:** (*Panicum maximum*)
 - Very common, grows well in shaded areas.
 3. **Giant Star Grass:** (*Cynodon nlemfuensis*)
 - A "creeping" grass that covers the ground. Excellent for grazing as it is very tough and recovers quickly.

B. FORAGE LEGUMES (FOR PROTEIN)

1. **Centro:** (*Centrosema pubescens*)
 - A climbing legume, mixes well with grasses.
2. **Stylo:** (*Stylosanthes guianensis*)
 - A hardy, drought-resistant legume.
3. **Leucaena:** (*Leucaena leucocephala*)
 - A "browse tree." The protein-rich leaves are cut and fed to goats and cattle.

10.3 PASTURE CULTIVATION AND ESTABLISHMENT

- **Land Preparation:** The land must be cleared, ploughed, and harrowed just like for any other crop.
- **Planting:**
 - Grasses (like Elephant Grass) are planted using **cuttings (stems)**.
 - Legumes are usually planted using **seeds**.
- **Weed Control:** Weeds must be removed, as they compete with the pasture for nutrients.
- **Fertilizer Application:** Applying NPK fertilizer (or manure) will greatly increase the yield of the pasture.

10.4 FORAGE CONSERVATION (SAVING FOR THE DRY SEASON)

Forage is abundant in the wet season and very scarce in the dry season. Forage conservation is the critical process of **preserving wet-season excess for dry-season feeding**.

1. HAY

- **Definition:** Hay is forage that is preserved by **DRYING**.
- **Process:**
 1. Cut the grass/legumes at the right stage (before they flower).
 2. Spread them out in the sun to dry.
 3. Turn them regularly to ensure even-drying.
 4. The goal is to reduce the moisture content from 80% down to 15-20%.
 5. Once dry, it is "baled" (compressed) and stored in a "hay barn" (a dry, ventilated shed) to keep it out of the rain.
- **Principle:** Removing water stops all bacterial and fungal activity, so the grass does not rot.

2. SILAGE

- **Definition:** Silage is forage that is preserved by **ANAEROBIC FERMENTATION** (pickling).
- **Process:**
 1. Cut the forage (grass, maize) and let it "wilt" for a few hours (not dry).
 2. Chop the forage into very small pieces.
 3. Pack the chopped forage *tightly* into a "silo" (a pit, a tower, or a plastic bag).
 4. **Crucial Step:** It must be compacted to remove all **air (oxygen)**.
 5. It is then sealed to make it **airtight**.
- **Principle:** In the absence of oxygen (anaerobic), bacteria convert plant sugars into *lactic acid*. The acid "pickles" the grass, preserving it and giving it a sweet, sour smell.
- **Result:** A moist, high-energy feed.

Hay vs. Silage:

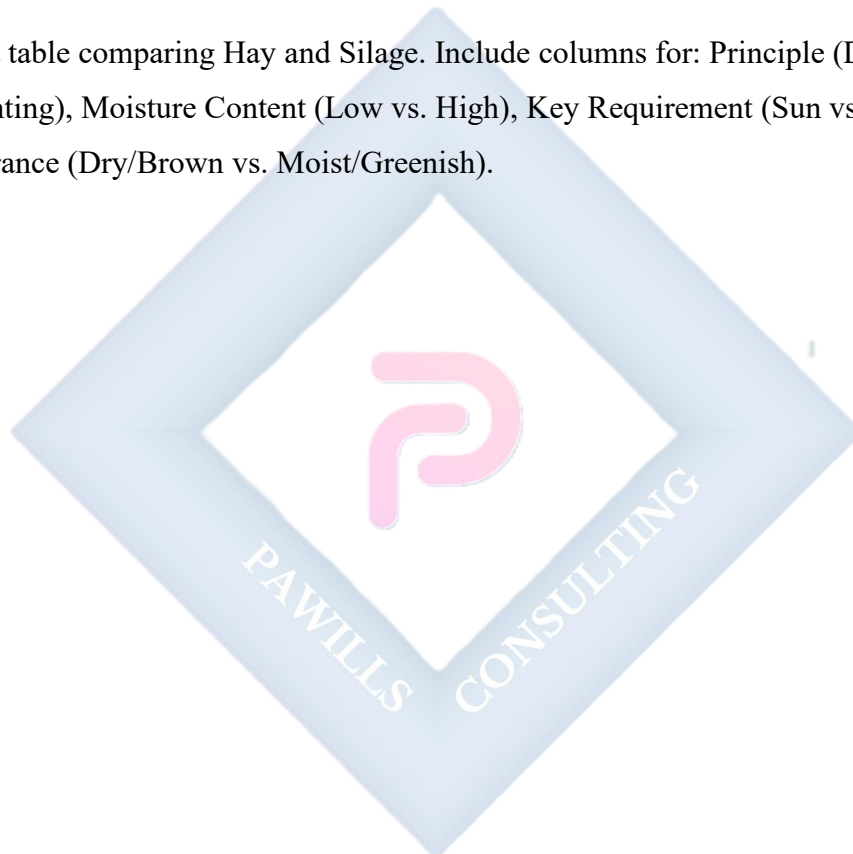
- **Hay** = Dried (like dry fish).
- **Silage** = Fermented (like ogi or pap).

EVALUATION (WEEK 10)

1. What is the main nutrient provided by grasses? What about legumes?
2. Name one common Forage Grass and one common Forage Legume.
3. Why is forage conservation so important in Nigeria?
4. Explain the main *principle* behind making Hay.
5. Explain the main *principle* behind making Silage. What is the one thing that *must* be removed during the process?

ASSIGNMENT (WEEK 10)

1. Draw a table comparing Hay and Silage. Include columns for: Principle (Drying vs. Fermenting), Moisture Content (Low vs. High), Key Requirement (Sun vs. Airtight), and Appearance (Dry/Brown vs. Moist/Greenish).



WEEK 11: REVISION

(This week is dedicated to a comprehensive review of all topics from Week 1 to Week 10 in preparation for the examination.)

- **Key Themes for Review:**

- Snail Farming (Housing, Feeding - Calcium, Breeding, Pests).
- Bee Keeping (Castes, Equipment - Smoker, Apiary Siting, Harvesting - Capped Honey, Extraction).
- Grasscutter Farming (Housing - Smooth Walls, Feeding - Elephant Grass, Breeding - Precocious Young).
- Animal Health I (Ecto vs. Endo parasites, examples, control methods).
- Animal Health II (Drugs vs. Vaccines, Biosecurity, Quarantine).
- Pasture (Grass vs. Legume, Hay vs. Silage).



SS2 LIVESTOCK FARMING LESSON NOTES

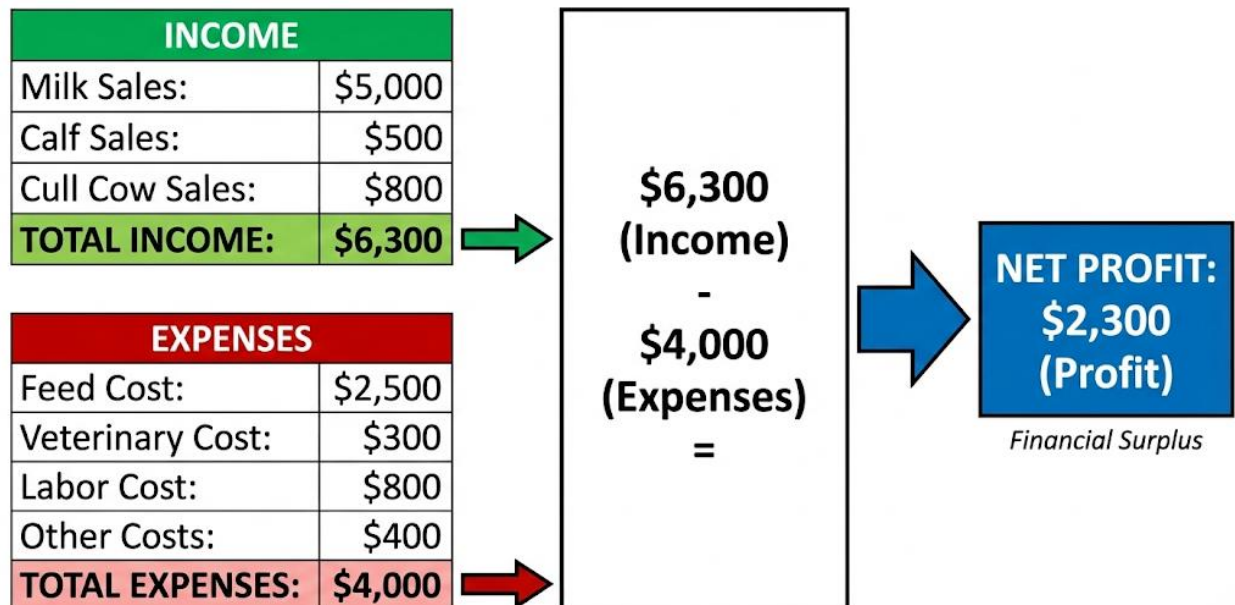
THIRD TERM

WEEK 1: INTRODUCTION TO LIVESTOCK ECONOMICS

1.1 PRINCIPLES OF FARM MANAGEMENT

Farm Management is the science and art of applying business principles to the running of a farm. It involves making decisions to organize and operate the farm (land, labour, capital, equipment) in a way that achieves the highest and most continuous profit.

LIVESTOCK ECONOMICS: ENTERPRISE BUDGET (Dairy Cow Unit)



A **Farm Manager** (which is you, the farmer) is a decision-maker who must perform several key functions:

1. **Planning:** Setting goals and objectives. (e.g., "My goal is to produce 1,000 table-size catfish in 6 months"). This includes planning for feed, medication, and sales.
2. **Organizing:** Assembling and coordinating all necessary resources (e.g., buying the fingerlings, purchasing feed, hiring labour).

3. **Directing (Implementation):** Carrying out the plan. This involves daily supervision of tasks like feeding, cleaning, and health checks.
4. **Controlling:** This is the "check and balance." It involves monitoring progress, comparing it to the plan, and making corrections. (e.g., "My mortality rate is 5% higher than planned, I must find out why").
5. **Coordinating:** Ensuring all parts of the farm (e.g., hatchery, grow-out, processing) work together smoothly.

Key Economic Principles in Livestock Farming:

- **Principle of Diminishing Returns:** As you add more of one input (e.g., feed) while holding others constant (e.g., pond size), the output (e.g., fish growth) will increase, but eventually, it will increase at a *decreasing* rate. Overfeeding past a certain point doesn't make the fish grow faster; it just wastes feed and pollutes the water.
- **Principle of Opportunity Cost:** The cost of choosing one option over another. The "opportunity cost" of spending N50,000 on snail farming is the profit you *could have* made if you had used that same N50,000 for poultry.
- **Principle of Substitution:** Replacing one input with another, cheaper one, as long as it gives the same result. (e.g., Replacing expensive Fish Meal with cheaper Soybean Meal in a feed formula).

1.2 ENTERPRISE BUDGETING

An **Enterprise** is a specific branch of the farm (e.g., the Poultry Enterprise, the Catfish Enterprise, the Snailery). A **Budget** is a written plan, in financial terms, that details the expected income and expected costs for an enterprise over a specific period. It is a "profit forecast."

Purpose of a Budget:

- To predict the profitability of an enterprise *before* you invest.
- To plan for financial needs (e.g., "I will need N200,000 in Month 3 for feed").
- To secure loans from banks (they *will* ask for your budget).

- To help in decision-making (e.g., "The budget shows snails are more profitable than quails").

Components of an Enterprise Budget:

1. **Revenue (Income):** The total money you expect to make.
 - *Formula:* Expected Yield (e.g., 950 kg of fish) x Expected Price (e.g., N1,500/kg) = Total Revenue (N1,425,000).
2. **Costs (Expenditure):** The total money you expect to spend. This is broken into two types:
 - **A) Variable Costs (Operating Costs):** Costs that *change* with the number of animals you raise. You *only* pay them if you are in production.
 - Examples: Feed, fingerlings/chicks, medication, casual labour, electricity/fuel, transport.
 - **Feed** is always the largest variable cost.
 - **B) Fixed Costs (Capital/Overhead Costs):** Costs that do *not* change, whether you raise 100 fish or 1,000 fish. These are the costs of *owning* the farm.
 - Examples: Rent/lease of land, depreciation of equipment (pump, generator), insurance, salary of permanent manager.
3. **Profit (or Loss):**
 - **Total Revenue - (Variable Costs + Fixed Costs) = Profit**

Simple Budget Example (100 Broilers for 8 weeks):

- **REVENUE:**
 - 100 broilers x 2.5kg avg. weight x N1,800/kg = **N450,000**
- **VARIABLE COSTS:**
 - 100 chicks @ N500/chick = N50,000
 - Feed (100 birds x 6kg/bird x N800/kg) = N480,000
 - Med/Vaccines = N10,000
 - Wood Shavings = N5,000
 - *Total Variable Costs* = N545,000

- **ANALYSIS:**

- Revenue (N450,000) - Variable Costs (N545,000) = **-N95,000 (Loss)**
- **Decision:** This budget immediately shows a massive loss. The farmer must find a way to reduce the feed cost or increase the sale price, or *not start the business*. This is the power of budgeting.

EVALUATION (WEEK 1)

1. Define Farm Management and list its five key functions.
2. Explain the "Principle of Diminishing Returns" using feed as an example.
3. What is the difference between a "Variable Cost" and a "Fixed Cost"? Give two examples of each.
4. Why is an enterprise budget a crucial tool for a farmer *before* starting production?
5. What is the single largest variable cost in most intensive livestock farms?

ASSIGNMENT (WEEK 1)

1. Create a simple enterprise budget for a 6-month Catfish grow-out cycle.
 - **Assumptions:** 200 fingerlings, 90% survival, 1kg avg. weight, N1,500/kg sale price.
 - **Costs:** Fingerlings (N50 each), Feed (N400,000 total), Meds/Salt (N10,000), Labour (N20,000).
 - Calculate the Total Revenue, Total Variable Costs, and the final Profit (or Loss).

WEEK 2: FARM RECORDS AND ACCOUNTS

2.1 INTRODUCTION

If budgeting is "predicting the future," record-keeping is "understanding the past." **Farm records are the eyes and ears of the farm manager.** A farmer who does not keep records is farming blindly.



2.2 IMPORTANCE OF RECORDS

1. **Performance Tracking:** To know if your farm is performing well. (e.g., "My FCR this cycle was 1.2:1, which is better than last cycle's 1.5:1").
2. **Profit and Loss Calculation:** To know *for a fact* if you are making money or losing money.
3. **Access to Credit:** Banks and investors will *not* give you a loan without seeing your records.
4. **Decision Making:** Records provide the *data* for good decisions. (e.g., "Record shows my birds always get sick in Week 3, I should take preventive action").

5. **Efficiency and FCR:** You cannot calculate your Food Conversion Ratio (FCR) without feed and weight records.
6. **Health Management:** Tracking vaccination dates, treatments, and mortality helps you spot and solve health problems.
7. **Planning and Budgeting:** Past records are the *best* source of information for creating accurate future budgets.

2.3 TYPES OF FARM RECORDS

A. PRODUCTION RECORDS

These track the physical performance of the livestock.

- **Poultry (Layers):**
 - Daily egg production (total eggs, broken, and % lay).
 - Daily mortality.
 - Daily feed consumption.
- **Poultry (Broilers):**
 - Weekly weight gain (by sampling).
 - Daily mortality.
 - Daily feed consumption.
- **Fish (Aquaculture):**
 - Stocking details (date, number, average weight).
 - Daily feed consumption.
 - Daily mortality.
 - Water quality readings (pH, temperature).
- **Ruminants (Goats/Cattle):**
 - Breeding records (date mated, sire, due date).
 - Birth records (date, number of kids, birth weight).
 - Milk production (for dairy).

B. HEALTH RECORDS

- **Vaccination Record:** Date, type of vaccine, brand, batch number.
- **Treatment Record:** Date, animal ID, symptoms, drug used, dosage, withdrawal period.
- **Mortality Record:** Date, number of deaths, and (if known) cause of death.

C. INVENTORY RECORDS

A list of all farm assets and supplies.

- **Assets:** Buildings, land, generator, vehicles, pumps, cages.
- **Supplies (Consumables):** Feed in store (bags), drugs/vaccines in store, fuel.

D. FINANCIAL RECORDS (FARM ACCOUNTS)

These track all the money coming in and going out.

1. **Farm Cash Book / Ledger:** The most basic and essential record. A simple T-account or spreadsheet showing:
 - **Date | Description | Income (Revenue) | Expenditure (Cost) | Balance**
2. **Profit and Loss (P&L) Statement:** A summary (usually at the end of a cycle or year) that shows:
 - **Total Revenue - Total Costs = Net Profit (or Loss)**
3. **Balance Sheet:** A "snapshot" of the farm's financial health on a specific day.
 - **Assets** (What you OWN: e.g., cash, animals, buildings) = **Liabilities** (What you OWE: e.g., loans) + **Equity** (Your net worth).

2.4 PRACTICAL RECORD KEEPING (EXAMPLES)

Example: Simple Poultry Daily Record (Pen A)

Date	No. of Birds	Feed (kg)	Mortality	Eggs	Remarks
Oct 28	500	50	1	380	Watery droppings in 2 birds
Oct 29	499	50	0	385	Treated with antibiotics
Oct 30	499	50	0	390	Droppings look better

Example: Simple Farm Cash Book

Date	Description	Income (₦)	Expenditure (₦)	Balance (₦)
Nov 1	Starting Balance			50,000
Nov 1	Bought 10 bags of feed		80,000	-30,000
Nov 2	Sold 50 crates of eggs	200,000		170,000
Nov 3	Paid labour (Mr. John)		15,000	155,000

EVALUATION (WEEK 2)

1. State five key reasons why farm records are essential for a livestock farmer.
2. What is the difference between a "Production Record" and a "Health Record"?
3. What is an "Inventory Record" used for?
4. What is a "Profit and Loss Statement"?
5. What is the "withdrawal period" on a health record, and why is it important?

ASSIGNMENT (WEEK 2)

1. Design a simple, one-page "Snailery Pen Record" sheet. Include columns you think are important for tracking the performance of snails.
2. You are a farmer. You start the week with N20,000. You buy 2 bags of feed for N8,500 each. You buy drugs for N1,500. You sell a goat for N35,000. Draw a simple Farm Cash Book for the week and show the final balance.

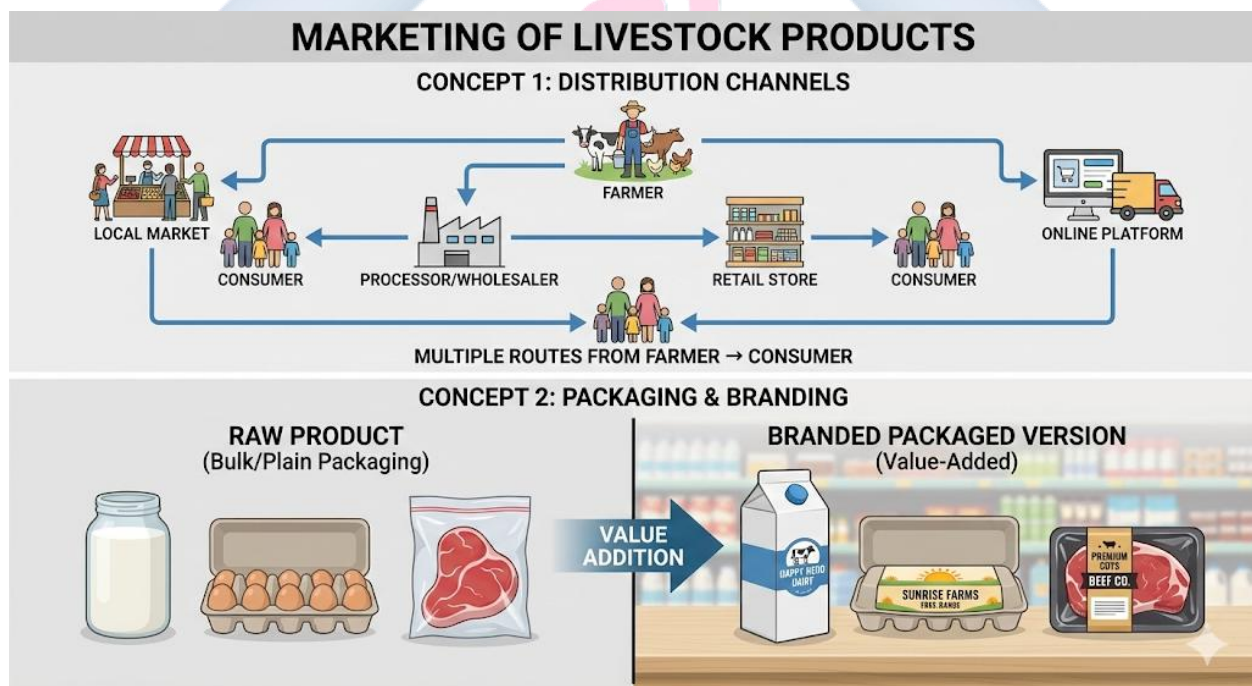
WEEK 3: MARKETING OF LIVESTOCK PRODUCTS I

3.1 DEFINITION OF MARKETING

Marketing is *not* just "selling." Selling is the final step. Marketing is the entire process of identifying what the customer wants and getting your product from the farm to that customer, at the right price, in the right form, and at the right time.

It includes:

- Market research (What do people want?)
- Production (Growing the product)
- Processing & Packaging (Adding value)
- Storage & Transport (Logistics)
- Pricing & Promotion (Advertising)
- Sales



3.2 CHANNELS OF DISTRIBUTION

This refers to the "path" or "chain" that a product takes from the producer (farmer) to the final consumer.

1. Zero-Level Channel (Direct Marketing):

- **Path:** Farmer -> Consumer
- **Example:** Farm-gate sales, farmer's markets, "point-and-kill" restaurants.
- *Pros:* Highest profit margin (no middleman), immediate cash.
- *Cons:* Small volume, time-consuming, farmer handles all marketing.

2. One-Level Channel (Retail Channel):

- **Path:** Farmer -> Retailer (e.g., local butcher, market woman) -> Consumer
- **Example:** You sell 100 chickens to Mrs. B, who sells them one by one in the market.
- *Pros:* Simple, farmer sells in bulk.
- *Cons:* Retailer takes a cut of the profit, may have low storage.

3. Two-Level Channel (Wholesale Channel):

- **Path:** Farmer → Wholesaler → Retailer → Consumer
- **Example:** You sell 1,000kg of fish to a "Wholesaler" (who has a cold room), who then sells in 100kg batches to 10 "Retailers," who then sell to consumers.
- *Pros:* Farmer sells *very* large volumes in one transaction.
- *Cons:* Farmer gets the lowest price per unit; has little control over the final product.

4. Three-Level Channel (Processor Channel):

- **Path:** Farmer → Processor → Wholesaler → Retailer → Consumer
- **Example:** You sell your cattle to a meat processing company. They turn it into sausages. The sausages are sold to a distributor (wholesaler), who sells to supermarkets (retailers).
- *Pros:* Guarantees a market for raw materials.
- *Cons:* Farmer is just one small part of a long chain.

3.3 PRICING OF LIVESTOCK PRODUCTS

Setting the right price is crucial. Too high, and no one buys. Too low, and you lose money.

Common Pricing Strategies:

1. Cost-Plus Pricing (The Farmer's Method):

- This is the most basic and safest method.
- **Formula:** Total Cost of Production + Profit Margin = Selling Price
- *Example:* It cost you N2,000 to raise one broiler. You want a 25% profit margin (N500). Your selling price must be *at least* N2,500.

2. Market-Based Pricing (Competitor Pricing):

- Setting your price based on what your competitors are charging for a similar product.
- *Example:* If all other farmers are selling eggs for N2,500/crate, you must price yours near N2,500, regardless of your cost.
- *Problem:* If your production cost is N2,800, but the market price is N2,500, you will lose money.

3. Value-Based Pricing (Premium Pricing):

- Setting a high price because the customer perceives your product as high-quality.
- *Example:* Selling "Organically-Fed, Free-Range" chicken for N5,000 when the market price is N3,000. This requires strong branding and packaging.

3.4 MARKET SURVEYS

A market survey is the process of gathering information to understand your target market. It's "doing your homework" before you start.

• What to Ask:

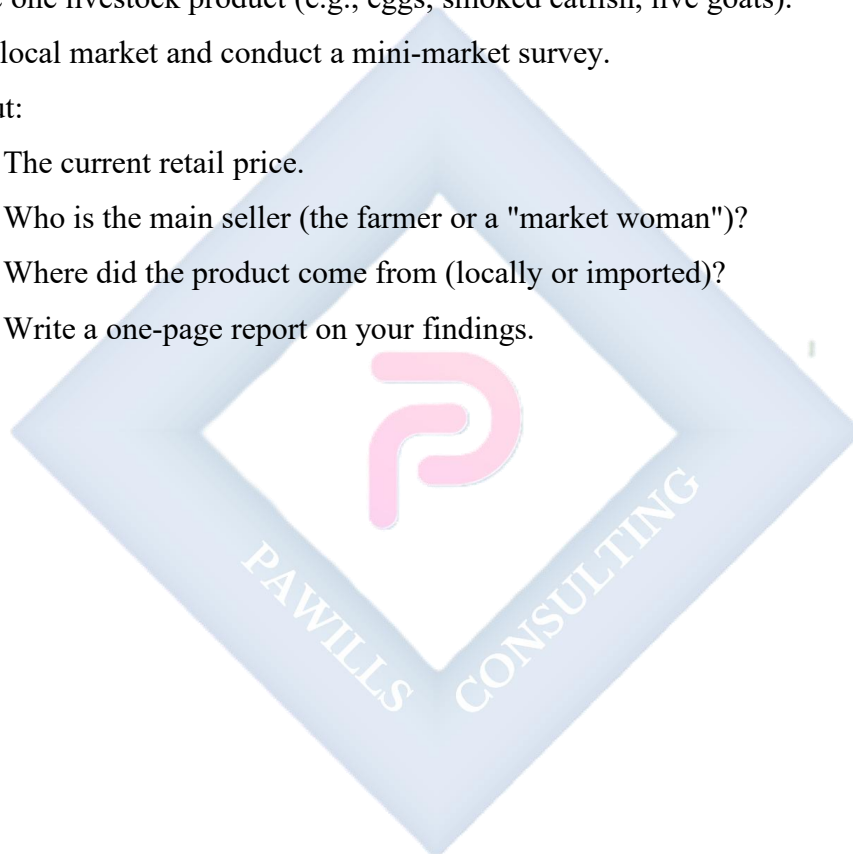
- *Who* will buy my product? (Hotels, market women, families).
- *What* form do they want it in? (Live, smoked, frozen).
- *What* size do they prefer? (e.g., "point-and-kill" buyers want 1kg+ fish).
- *Where* are they? (Is the market 5km away or 50km away?).
- *When* do they buy? (e.g., Demand for rams peaks at Sallah).
- *What* price are they willing to pay?

EVALUATION (WEEK 3)

1. Define "Marketing" and explain why it is more than just "selling."
2. Draw a diagram of a "Two-Level Channel" of distribution.
3. What are the pros and cons of "Direct Marketing" (Zero-Level Channel)?
4. Explain "Cost-Plus Pricing."
5. Why is it important to conduct a market survey *before* starting your farm?

ASSIGNMENT (WEEK 3)

1. Choose one livestock product (e.g., eggs, smoked catfish, live goats).
2. Visit a local market and conduct a mini-market survey.
3. Find out:
 - The current retail price.
 - Who is the main seller (the farmer or a "market woman")?
 - Where did the product come from (locally or imported)?
 - Write a one-page report on your findings.



WEEK 4: MARKETING OF LIVESTOCK PRODUCTS II

4.1 PACKAGING

Packaging is the science and art of enclosing or protecting a product for transport, storage, sale, and use. Good packaging is a key marketing tool.

Functions of Packaging:

1. **Protection and Preservation:** This is the primary function.
 - It protects the product from physical damage (e.g., egg cartons protect eggs from breaking).
 - It protects from contamination (dust, germs).
 - It protects from spoilage (e.g., vacuum-sealing meat removes oxygen and extends shelf life).
2. **Information (Labeling):** The package *must* legally provide information to the consumer.
 - Product name ("Fresh Chicken")
 - Net weight
 - Expiry date / Best-before date
 - NAFDAC number (if processed)
 - Storage instructions ("Keep frozen")
3. **Marketing and Appeal (Branding):**
 - The package is the "silent salesman." A well-designed, attractive package will catch the customer's eye.
 - It communicates the brand's identity (e.g., "premium," "organic," "local").
4. **Convenience and Containment:**
 - It makes the product easy to handle, transport, and store (e.g., a 1-litre milk carton, a crate of eggs).

Packaging Examples:

- **Eggs:** Paper/plastic crates.
- **Milk:** Plastic bottles, Tetra Paks.

- **Meat (Fresh):** Styrofoam trays with plastic wrap, vacuum-sealed bags.
- **Smoked Fish:** Cardboard cartons, plastic-wrapped.

4.2 BRANDING

Branding is the process of creating a unique name, logo, and identity for your product to differentiate it from competitors. A brand is a "promise" of quality.

- **Brand Name:** The name of your product (e.g., "Adebayo Farms," "Chi Chicken").
- **Logo:** A unique symbol or design for your brand.
- **Slogan:** A catchy phrase ("Always fresh, always tasty").
- **Reputation:** The most important part. A brand is built on a *consistent* reputation for quality.

Why is Branding Important?

1. **Recognition:** It helps customers find your product easily.
2. **Trust:** Customers are more likely to buy a branded product they trust, even if it's new.
3. **Loyalty:** A good brand creates loyal customers who will always buy from you.
4. **Premium Pricing:** A strong brand can charge a *higher price* than a generic, unbranded product. (e.g., People will pay more for "Chi" chicken than for an unlabelled bag of chicken).

4.3 EXPORT OPPORTUNITIES

There is a significant international market for Nigerian livestock products, especially those that are unique to West Africa.

- **Products in Demand:**
 - **Snail Meat:** (Frozen or canned). High demand in Europe and USA.
 - **Smoked/Dried Fish:** (Catfish, Bonga fish). High demand from the African diaspora.
 - **Hides and Skins:** (Especially "Kano Brown" and "Sokoto Red" goat skins) are world-famous for their quality (Moroccan leather).

- **Challenges of Export:** The export market is *very* difficult to enter due to high standards.
 - **Sanitary and Phytosanitary Standards (SPS):** These are international rules to protect human and animal health.
 - **Quality & Consistency:** The product must be perfect, every time.
 - **Processing & Packaging:** Must meet international standards (e.g., vacuum sealing, blast freezing, proper labeling).
 - **Certifications:** You need approvals from bodies like NAFDAC, Nigerian Export Promotion Council (NEPC), and the quarantine service (NAQS).

EVALUATION (WEEK 4)

1. List the four main functions of packaging.
2. Why is an egg carton a perfect example of good packaging?
3. Define "Branding." Why is it important for a farmer?
4. What is the difference between a Brand Name and a Logo?
5. List two Nigerian livestock products with high export potential and two major challenges in exporting them.

ASSIGNMENT (WEEK 4)

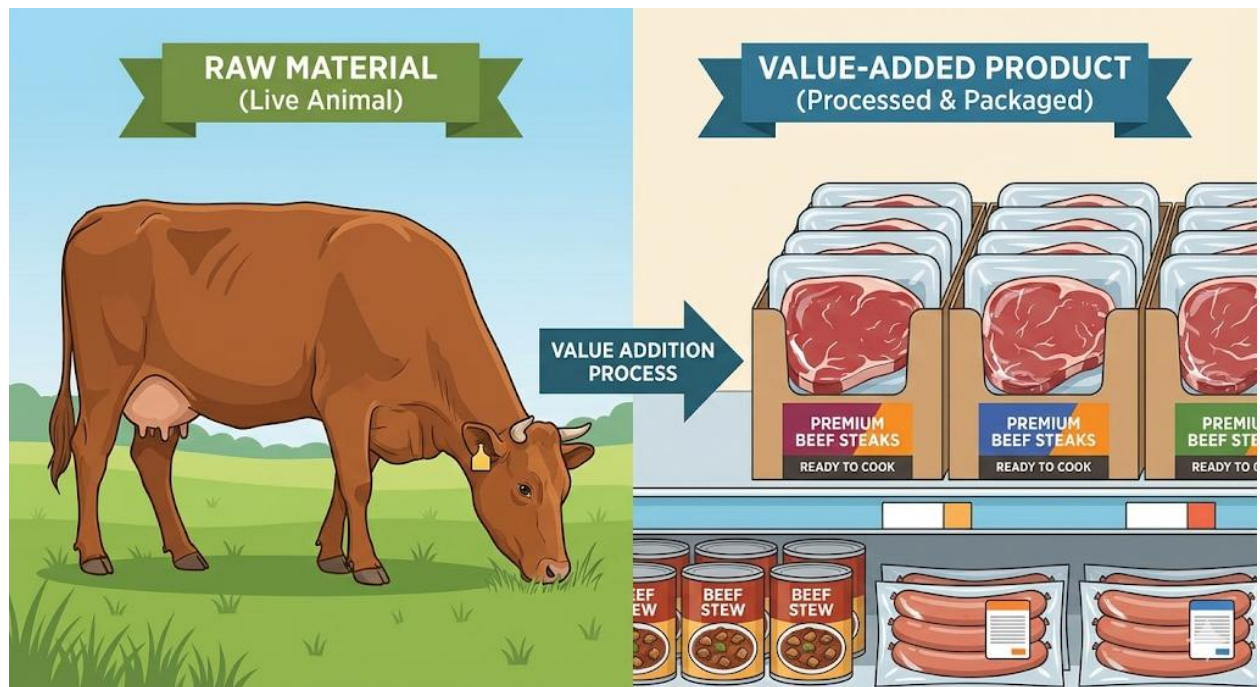
1. Go to a supermarket or local shop. Find two competing brands of the *same* product (e.g., two different brands of milk, or eggs).
2. Compare their packaging and branding.
3. Write a short report answering:
 - Which one has a more attractive package? Why?
 - What information is on the label?
 - Which one is more expensive?
 - Which one would you buy, and why?

WEEK 5: MIDTERM TEST

(This week is for assessment of all topics covered in Weeks 1-4: Economics, Records, Marketing I, and Marketing II.)



WEEK 6: VALUE ADDITION IN LIVESTOCK



6.1 WHAT IS VALUE ADDITION?

Value Addition (or Processing) is the process of taking a raw livestock product (e.g., live cow, raw milk) and changing its form to create a new, more valuable product (e.g., sausages, yoghurt).

Why Add Value?

1. **Higher Profit:** The processed product almost *always* sells for more money than the raw product.
2. **Longer Shelf Life:** Processing (like smoking, drying, pasteurizing) preserves the food, stopping it from spoiling. A live chicken must be sold in days; a frozen chicken lasts for months.
3. **New Markets:** It opens up new customers (e.g., supermarkets, exporters).
4. **Reduced Waste:** It allows you to use 100% of the animal.
5. **Product Differentiation:** It makes your product unique.

6.2 PROCESSING MEAT

- **Fresh Meat:** Hygienic slaughtering in an abattoir, cutting into specific "cuts" (steak, brisket), and packaging (e.g., vacuum-sealing).
- **Processed Products:**
 - **Smoked:** (e.g., Smoked chicken, smoked fish).
 - **Grilled:** (e.g., Suya, Asun).
 - **Dried:** (e.g., Kilishi, Biltong).
 - **Ground/Minced:** (e.g., Burger patties, minced meat).
 - **Cured/Cased:** (e.g., Sausages, hot dogs, bacon).

6.3 PROCESSING MILK (DAIRY)

Raw milk spoils in hours. Processed milk lasts for days, weeks, or months.

- **Pasteurization:** Heating milk briefly (e.g., 72°C for 15 seconds) to kill harmful bacteria and extend its shelf life.
- **Processed Products:**
 - **Yoghurt:** Fermented milk.
 - **Cheese:** (e.g., "Wara" - local soft cheese).
 - **Butter:** Churning the fat from milk.
 - **"Fura da Nono":** Fermented milk, often mixed with millet.
 - **Evaporated Milk / Powdered Milk:** (Industrial processing).

6.4 PROCESSING HIDES AND SKINS

- **Raw Products:** Hides (from cattle) and Skins (from goats/sheep).
- **Value Addition Process (Curing):** This is the first step to stop the skin from rotting *before* it gets to the tannery.
 - **Flaying:** Skillful removal of the skin from the carcass.
 - **Salting:** Rubbing large amounts of salt onto the flesh side to draw out water.
 - **Sun Drying:** Stretching the skin on a frame to dry in the sun.
- **Final Product:** The cured skins are sold to **Tanneries**, which turn them into **Leather**.

6.5 UTILIZATION OF BY-PRODUCTS ("ZERO WASTE" FARMING)

By-products are the "leftovers" from processing. A smart farmer sees these as another source of income.

- **Manure (Droppings):**
 - *Value Added:* Sold as high-quality organic fertilizer.
 - *Value Added:* Can be used in a "biogas digester" to create cooking gas.
- **Bones:**
 - *Value Added:* Cooked, dried, and ground into **Bone Meal**, a rich calcium supplement for animal feed.
- **Blood:**
 - *Value Added:* Cooked, dried, and ground into **Blood Meal**, a very high-protein supplement for feed.
- **Feathers (from poultry):**
 - *Value Added:* Processed into **Feather Meal** for feed.
- **Offal (Intestines, liver, heart, etc.):**
 - *Value Added:* Cleaned and sold in the local market as "assorted meat."

EVALUATION (WEEK 6)

1. Define "Value Addition."
2. Give three reasons why a farmer should add value to their products.
3. List three processed products from meat and three from milk.
4. What is "Curing" in relation to hides and skins, and why is it necessary?
5. What are "by-products"? Give three examples of livestock by-products and their value-added use.

ASSIGNMENT (WEEK 6)

1. Choose one value-added process (e.g., making Wara from milk, or Kilishi from beef).
2. Research the steps involved.
3. Create a simple flowchart showing the process from the raw material to the final product.

WEEK 7: MIDTERM BREAK

(No new lesson notes for this week.)



WEEK 8: COOPERATIVE SOCIETIES IN LIVESTOCK

8.1 DEFINITION AND NATURE

A **Cooperative Society** is a business organization owned and democratically controlled by its members. It is a group of people (e.g., farmers) who voluntarily join together to pool their resources and efforts to achieve a common economic goal.

- **Motto:** "Each for all, and all for each."
- **Principle:** It is run for the *benefit of the members*, not for outside investors. Profits are often shared back to members as "dividends."



8.2 ROLES AND FUNCTIONS OF LIVESTOCK COOPERATIVES

1. Bulk Purchasing (Input Supply):

- A single farmer buys 10 bags of feed. A cooperative of 100 farmers buys 1,000 bags.
- The cooperative can buy directly from the manufacturer at a **much lower wholesale price**.

- This (feed, drugs, fingerlings) is one of the biggest benefits.
- 2. **Bulk Marketing:**
 - A single farmer has 500kg of fish to sell, giving him weak bargaining power.
 - A cooperative gathers all its members' fish (e.g., 10,000 kg) and negotiates a **high-volume contract** with a major hotel or processor. They have much stronger bargaining power.
- 3. **Access to Credit (Financial Services):**
 - The cooperative can act as a "micro-finance bank" or "Esusu/Ajo."
 - Members contribute to a central fund, from which they can take low-interest loans.
 - It is also *much* easier for a cooperative (as a legal group) to get a large loan from a commercial bank than for a single farmer.
- 4. **Shared Resources and Equipment:**
 - One farmer cannot afford a N10 million feed mill or a N5 million tractor.
 - A cooperative of 200 farmers can easily buy it *jointly*. The equipment is owned by the co-op and used by all members.
- 5. **Education and Extension:**
 - It is a perfect platform for education. An extension agent can teach 100 farmers at once during their co-op meeting, instead of visiting 100 individual farms.
 - Members can share knowledge and best practices.
- 6. **Political Lobbying:**
 - A single farmer's voice is small. A cooperative (representing 1,000 farmers) can speak to the government to lobby for local roads, electricity, or better policies.

8.3 BENEFITS OF COOPERATIVES TO FARMERS

- **Reduced Costs:** Cheaper inputs.
- **Higher Income:** Better prices for their products.
- **Access to Loans:** Easy and cheap credit.
- **Access to Technology:** Use of shared, modern equipment.
- **Knowledge:** Regular training and support.
- **Empowerment:** A stronger voice and less risk.

8.4 CHALLENGES OF COOPERATIVES

- **Mismanagement/Corruption:** Poor or dishonest leadership can steal funds.
- **Low Member Participation:** Members become inactive and don't attend meetings.
- **Loan Defaults:** Members take loans and refuse to pay back, crippling the co-op.
- **Internal Conflicts:** Disagreements between members.

EVALUATION (WEEK 8)

1. What is a Cooperative Society?
2. Explain "Bulk Purchasing" and why it is a benefit.
3. How do cooperatives help farmers get access to loans?
4. How do cooperatives help farmers "add value" or "share resources"?
5. List two common challenges that can cause a cooperative to fail.

ASSIGNMENT (WEEK 8)

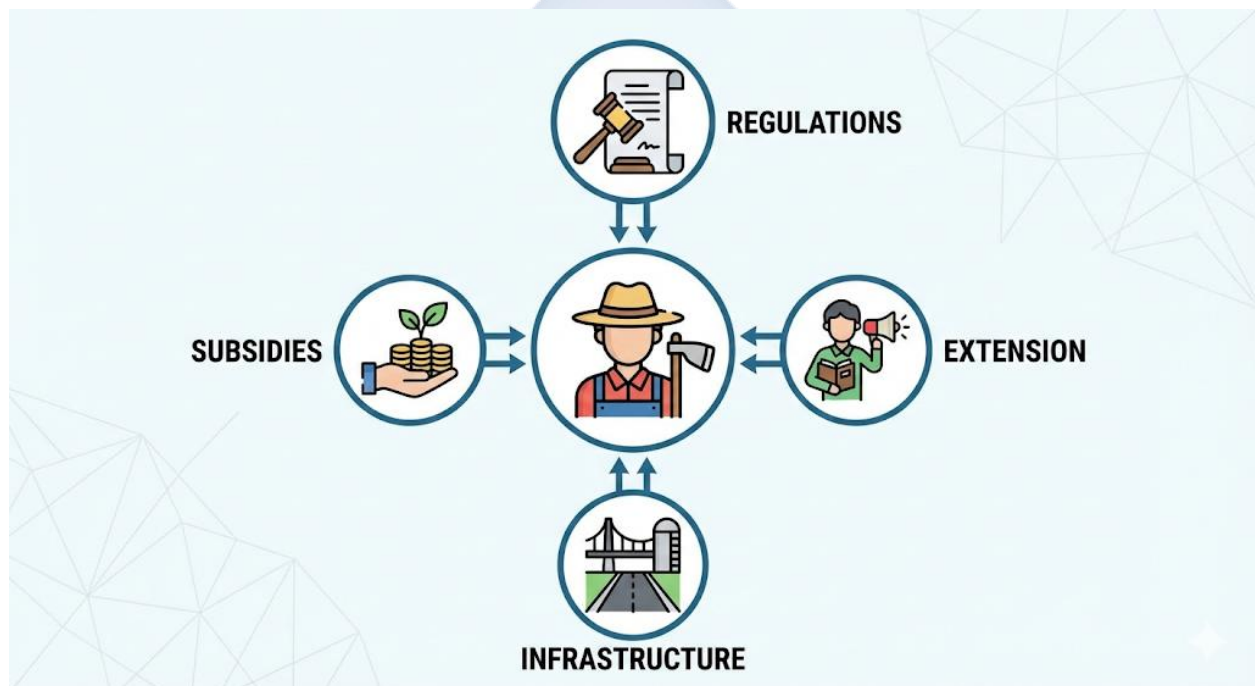
1. Find out if there is a "Farmers' Cooperative" or any other cooperative (e.g., "Butchers' Co-op," "Egg Sellers' Co-op") in your local government area.
2. Try to find out what their main function is (e.g., Do they give loans? Do they sell feed?).
Write a short report.

WEEK 9: GOVERNMENT POLICIES IN LIVESTOCK FARMING

9.1 ROLE OF GOVERNMENT

The government gets involved in agriculture and livestock farming because it is essential for:

- **National Food Security:** Ensuring the nation can feed itself.
- **Economic Growth:** Agriculture is a huge part of Nigeria's GDP.
- **Employment:** The sector employs over 50% of the population.
- **Public Health:** Ensuring food is safe and animal diseases don't spread to humans.



9.2 GOVERNMENT POLICIES AND PROGRAMMES

A. SUBSIDIES

- **Definition:** A subsidy is a payment (or discount) from the government to farmers to reduce their costs and encourage production.
- **Examples:**
 - **Input Subsidy:** Selling fertilizer, seeds, or fingerlings to farmers at a 50% discount.

- **Credit Subsidy:** The government partners with banks (e.g., CBN's Anchor Borrowers' Programme) to give farmers "soft loans" with very low-interest rates.
- **Price Support:** The government sets a "minimum guaranteed price" for a product (e.g., maize) to protect farmers from price collapse.

B. EXTENSION SERVICES

- **Definition:** This is the most important *educational* arm of the government. Extension Agents are the "teachers" who bridge the gap between agricultural research and the local farmer.
- **Role:**
 - To visit farms and teach farmers modern, scientific methods (e.g., "This is how you use a vaccine," "This is how you build a biosecure pen").
 - To introduce high-yield breeds and new technologies.
 - To collect data and report problems (e.g., "There is a disease outbreak in this area") from the farmers back to the research institutes.
- **Key Agency:** Agricultural Development Programmes (ADPs), one in every state.

C. REGULATIONS AND QUALITY CONTROL

- **Purpose:** To protect the public, the environment, and the national economy.
- **Key Agencies:**
 - **NAFDAC (National Agency for Food and Drug Administration and Control):** Regulates all *processed* food (e.g., milk, yoghurt, sausages, imported fish). They ensure food is safe for humans.
 - **SON (Standards Organisation of Nigeria):** Sets the quality standards for products, including equipment and some animal feeds.
 - **NAQS (Nigerian Agricultural Quarantine Service):** The "border police" for agriculture. They inspect animals, plants, and fish at all ports (sea, air, land) to prevent the importation or exportation of diseases.
 - **Veterinary Council of Nigeria (VCN):** Regulates the veterinary profession, ensuring only qualified vets practice.

D. INFRASTRUCTURE DEVELOPMENT

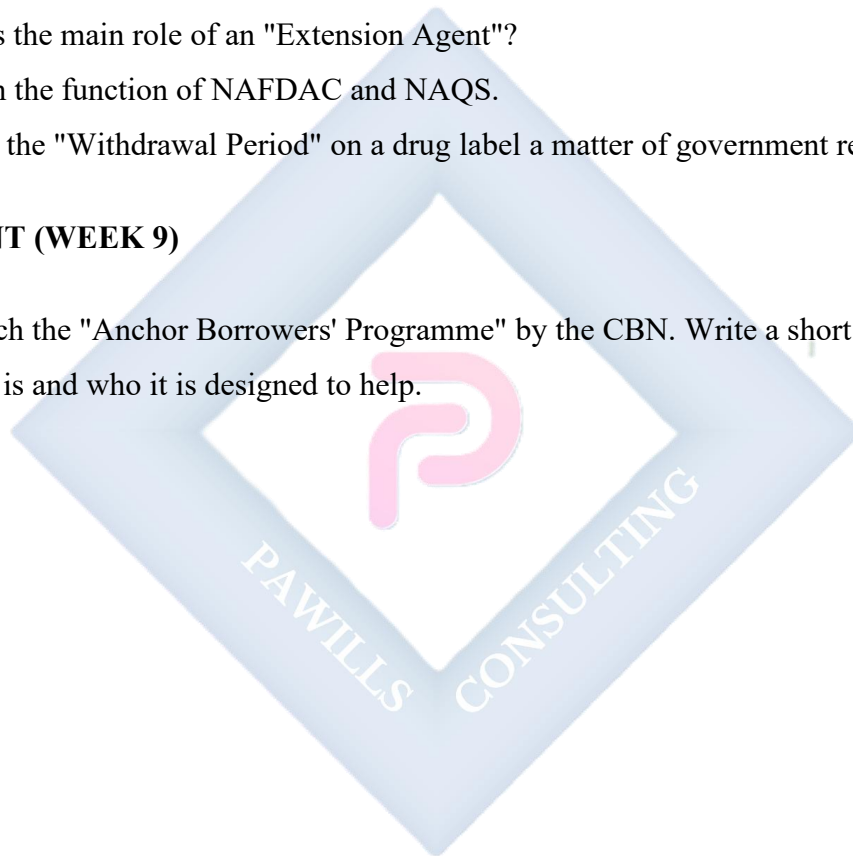
- The government is responsible for building public infrastructure that supports farming: rural roads (to get food to market), dams (for irrigation), and electricity (for processing).

EVALUATION (WEEK 9)

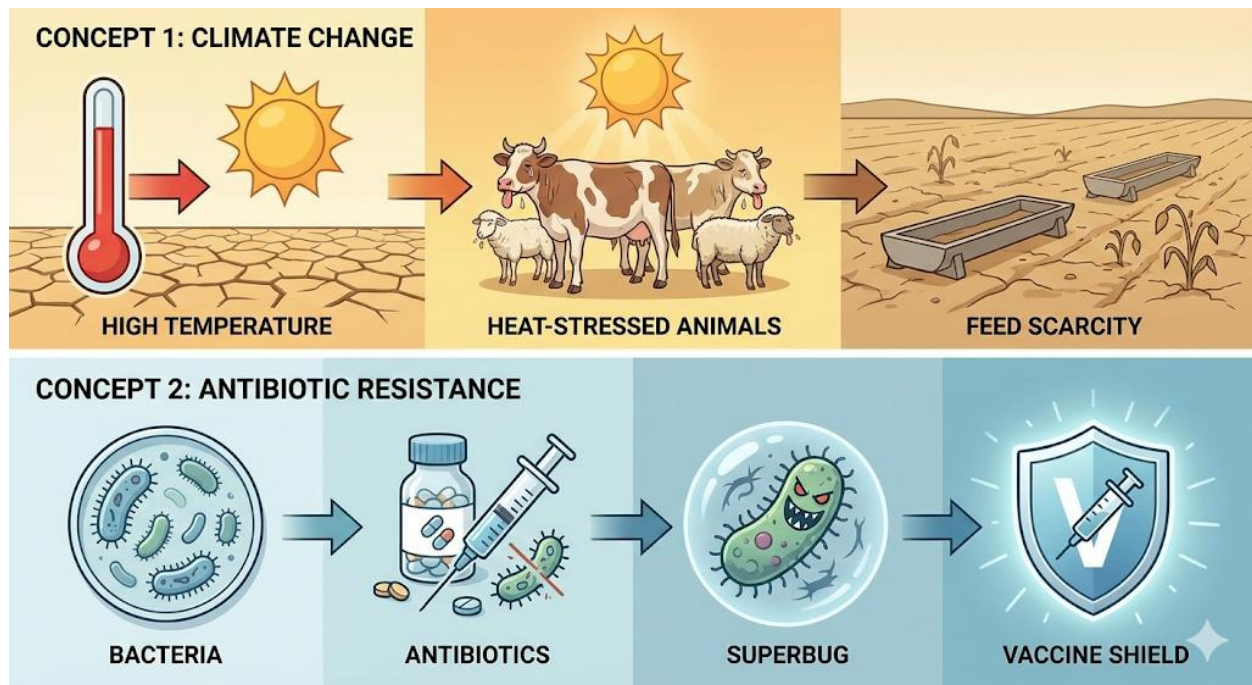
1. List three main reasons why the government is involved in livestock farming.
2. What is a "subsidy"? Give two examples of government subsidies for farmers.
3. What is the main role of an "Extension Agent"?
4. Explain the function of NAFDAC and NAQS.
5. Why is the "Withdrawal Period" on a drug label a matter of government regulation?

ASSIGNMENT (WEEK 9)

1. Research the "Anchor Borrowers' Programme" by the CBN. Write a short paragraph on what it is and who it is designed to help.



WEEK 10: CONTEMPORARY ISSUES IN LIVESTOCK FARMING



10.1 IMPACT OF CLIMATE CHANGE ON LIVESTOCK

Climate change (e.g., rising temperatures, unpredictable rainfall, drought) is a major threat to livestock farming.

- **1. Heat Stress:**
 - **Effect:** High temperatures (above 30°C) cause animals to "stress."
 - **Symptoms:** Animals eat less, drink more, and pant.
 - **Impact:**
 - Poultry: Lay fewer eggs, eggs have thin shells, broilers grow slower.
 - Cattle: Produce less milk, lower fertility.
 - High mortality (death).
- **2. Feed Scarcity:**
 - **Effect:** Droughts and poor rainfall lead to failed harvests of maize, soy, and grass.
 - **Impact:** The cost of feed (the #1 cost) increases dramatically. Pasture for ruminants dries up, leading to starvation and forced migration.
- **3. Water Scarcity:**

- **Effect:** Droughts dry up rivers and lower the water table.
- **Impact:** It becomes expensive to get water for fish ponds and for animals to drink.
- **4. Increased Disease:**
 - **Effect:** Flooding (from extreme rainfall) can contaminate ponds and water sources, spreading diseases like cholera. Changes in temperature also allow disease vectors (like ticks) to move into new areas.

Adaptation / Mitigation Strategies:

- Planting shade trees around pens.
- Better housing ventilation (cross-ventilation).
- Using heat-tolerant "indigenous" breeds.
- Forage conservation (making Hay and Silage) for the dry season.
- Water harvesting (building reservoirs).

10.2 BIOSECURITY MEASURES (AS A CONTEMPORARY ISSUE)

Biosecurity is no longer just "good practice"; it is an *essential* defence against modern, devastating diseases.

- **Why is it a "Contemporary Issue"?**
 - The increased global trade and movement of people and animals have made it easier for diseases to spread rapidly (e.t., Avian Influenza "Bird Flu," African Swine Fever).
 - These diseases can wipe out 100% of a flock and destroy a farmer's investment in days.
- **Biosecurity is a "Wall":**
 - It is a set of practices designed to prevent pathogens from *entering* the farm (the "wall") and from *spreading* within the farm.
- **Key Practices (Review):**
 - **Quarantine:** Isolating new animals.
 - **All-in, All-out:** Breaking the disease cycle.
 - **Footbaths & Disinfection:** For all visitors and vehicles.

- **Pest Control:** Fencing against rats, birds, and wild animals.
- **Cleanliness:** Regular disinfection of pens, feeders, and drinkers.

10.3 ANTIBIOTIC RESISTANCE

This is one of the biggest threats to global human and animal health.

- **The Problem:** For decades, farmers have used antibiotics *improperly*.
 - 1. As a "Growth Promoter": Adding low doses of antibiotics to animal feed *every day* to make them grow faster (this is now banned in many countries).
 - 2. As "Prophylaxis": Treating animals *before* they even get sick.
- **The Consequence:** This continuous, low-dose exposure has "trained" bacteria to become resistant. We have created "**Superbugs**"—bacteria that can no longer be killed by common antibiotics.
- **The Danger:**
 - The drugs no longer work when the animal *actually* gets sick.
 - These resistant bacteria can be passed to humans *through meat* or from farm workers, causing infections in people that are very difficult to treat.
- **The Solution:**
 - **Stop** using antibiotics as growth promoters.
 - Use antibiotics *only* to treat sick animals, as prescribed by a vet.
 - Focus on **Prevention:** Use good biosecurity, sanitation, and vaccines so the animals don't get sick in the first place.

EVALUATION (WEEK 10)

1. Explain two ways climate change (heat stress) directly impacts a poultry farmer.
2. How does climate change lead to scarcity of pasture and feed?
3. Why is biosecurity more important today than ever before?

4. What is "Antibiotic Resistance"?
5. Explain the *improper* use of antibiotics that leads to resistance, and state the correct solution.

ASSIGNMENT (WEEK 10)

1. Imagine you are a livestock extension agent. Write a short flyer (one page) for local farmers explaining three "Climate Change Adaptation Strategies" they can use on their farms (e.g., building shade, conserving forage).



WEEK 11: REVISION

(This week is dedicated to a comprehensive review of all topics from Week 1 to Week 10 in preparation for the examination.)

- **Key Themes for Review:**

- Farm Management & Economics (Budgeting, Variable/Fixed Costs).
- Record Keeping (Production, Health, Financial).
- Marketing (Channels, Pricing, Packaging, Branding).
- Value Addition (Meat, Milk, Hides, By-products).
- Cooperatives (Bulk purchase, Credit, Shared resources).
- Government Policies (Subsidies, Extension, Regulations - NAFDAC/NAQS).
- Contemporary Issues (Climate Change, Biosecurity, Antibiotic Resistance).

